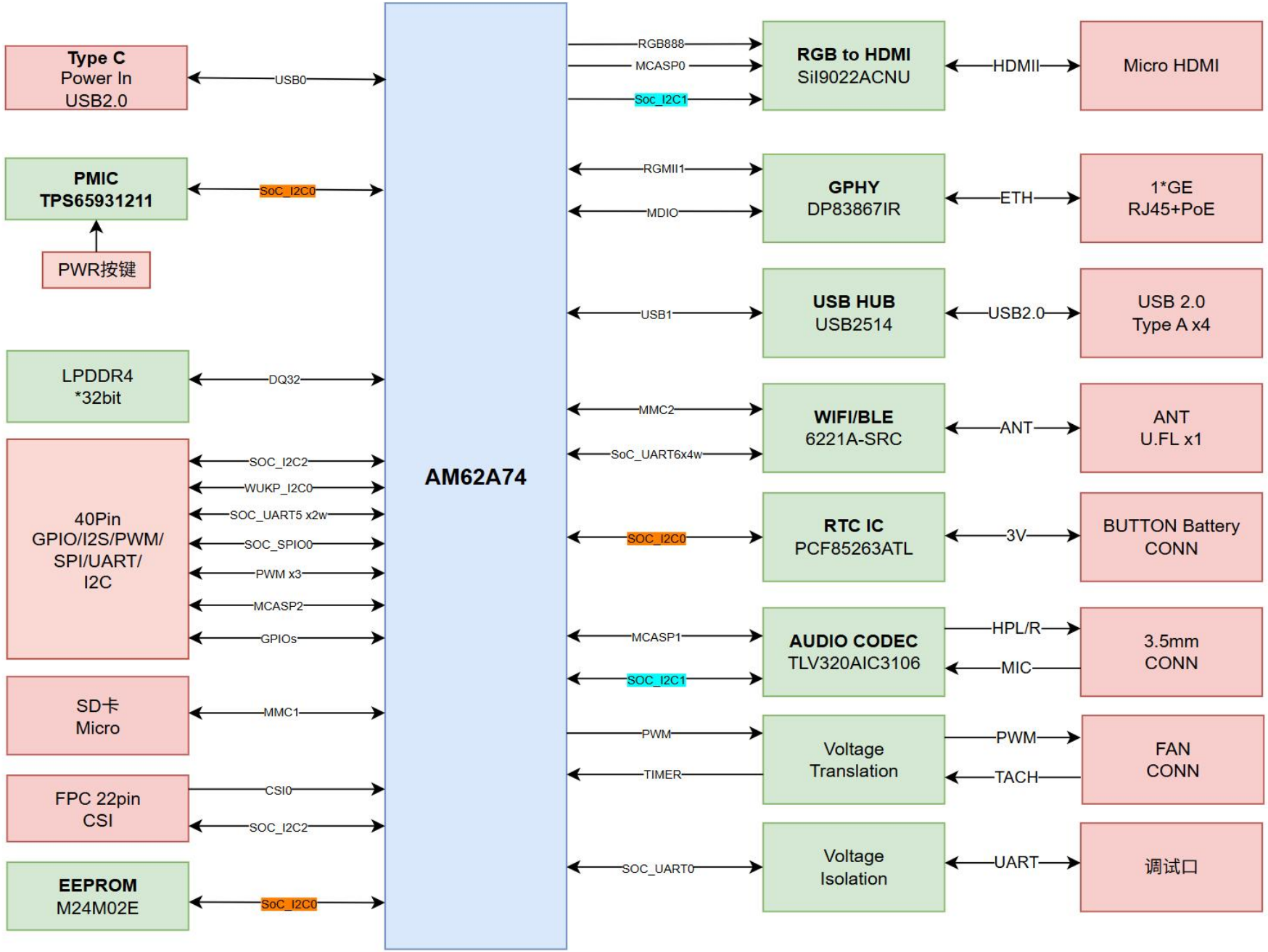
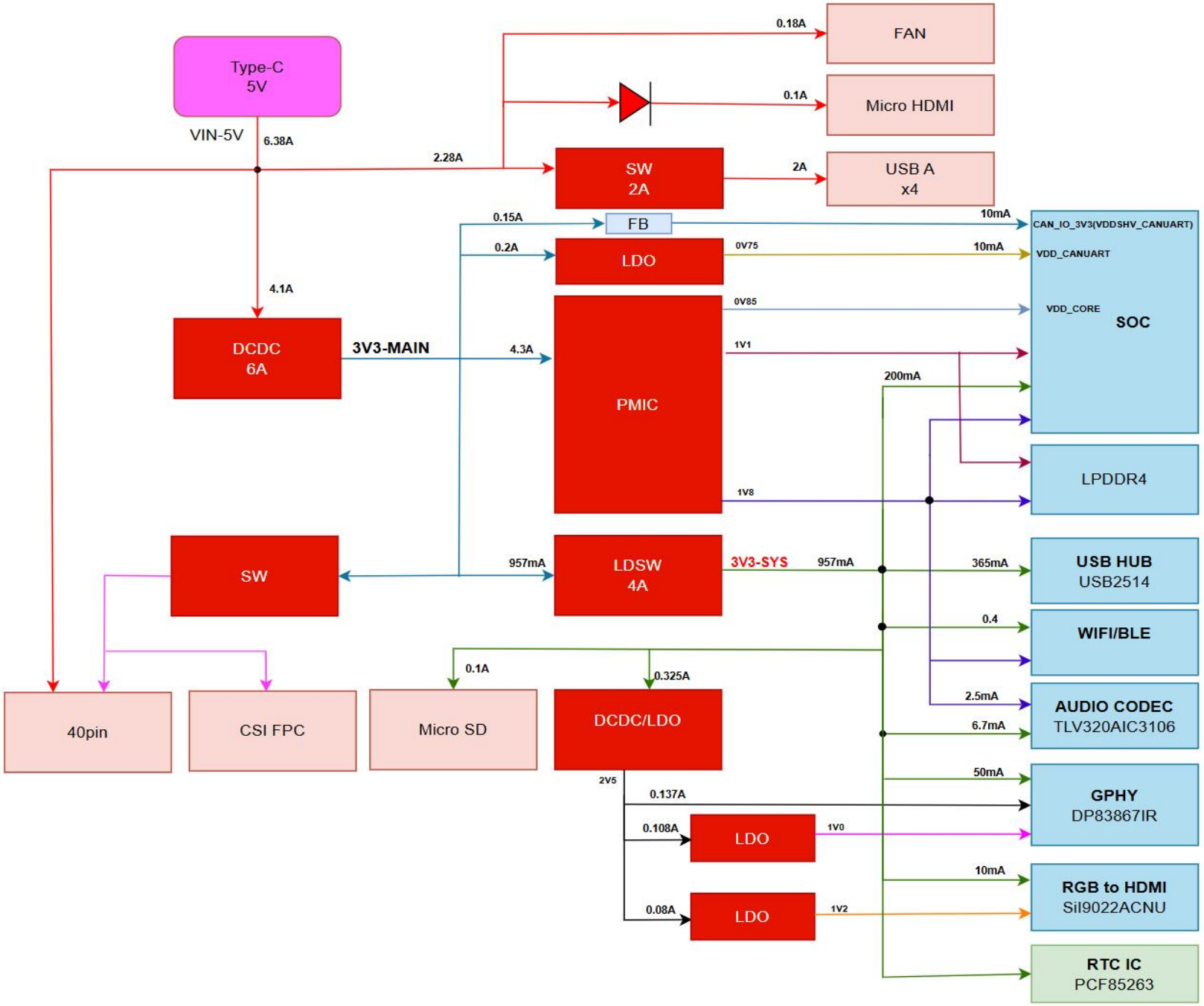
BLOCK DIAGRAM AM62A



<Core Design>

inhand	Beijing InHand Networks Technology Co., Ltd.				
Project:	Mo_62A				
File:	01. BLOCK DGM AM62A				
Date:	Monday, April 27, 2026			Rev:	V1.0
Designed by:		Reviewed by:		Sheet:	1 of 22

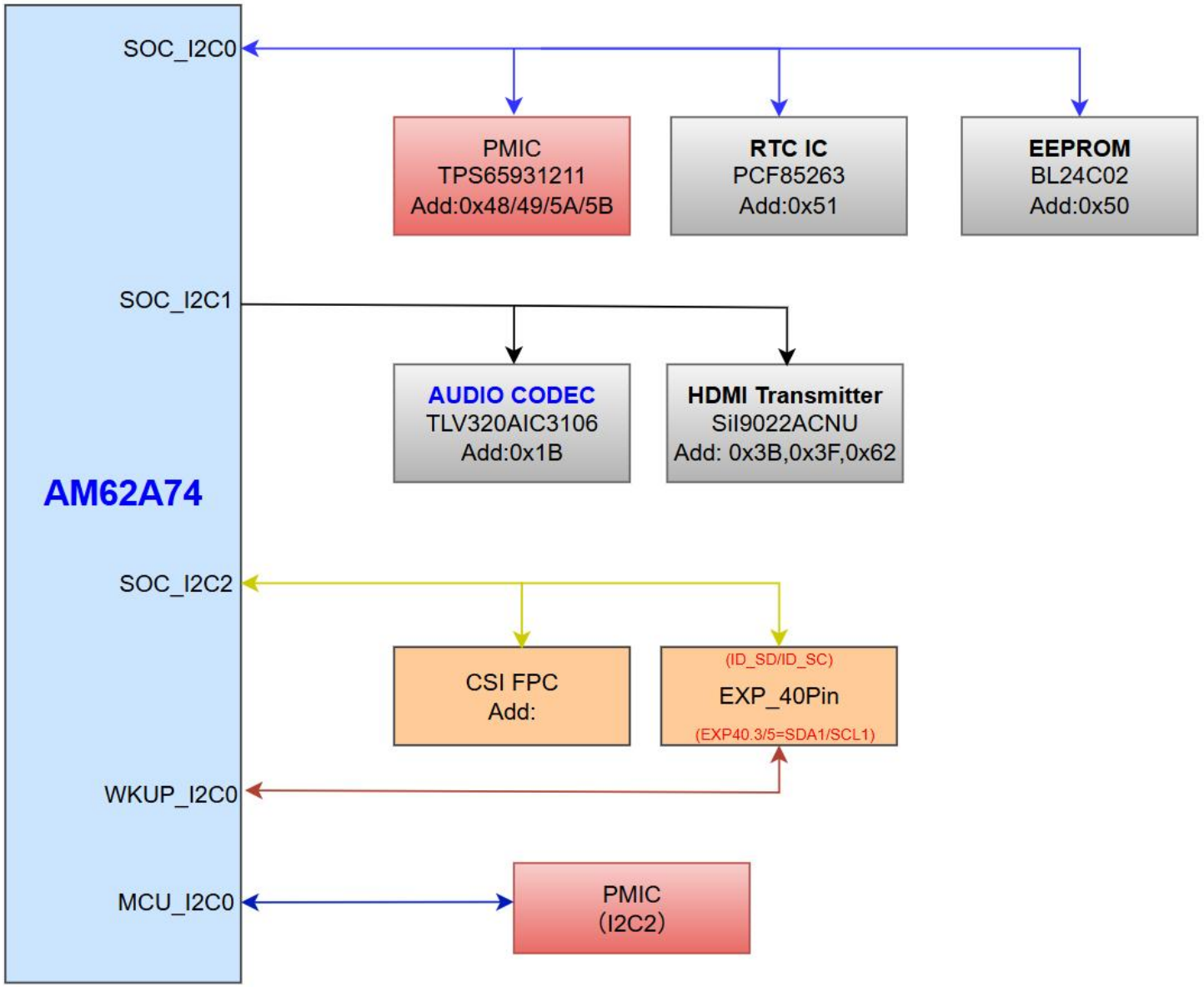
POWER BLOCK DGM



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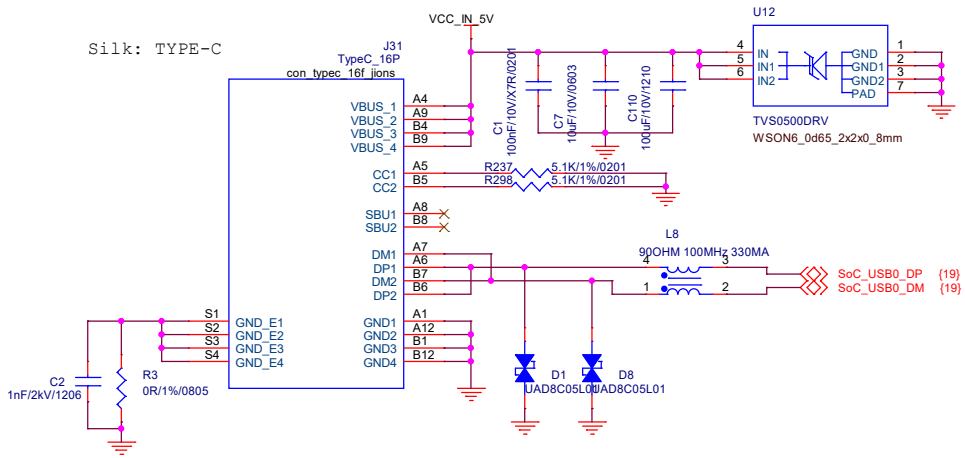
inhand	Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A			
File:	02. POWER BLOCK DGM			
Date:	Sunday, April 26, 2026		Rev:	V1.0
Designed by:		Reviewed by:	Sheet:	2 of 22

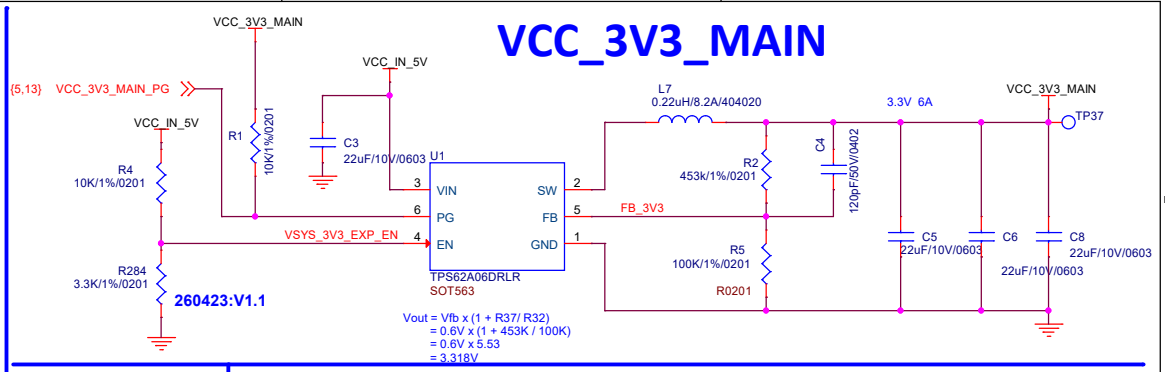
I2C TREE



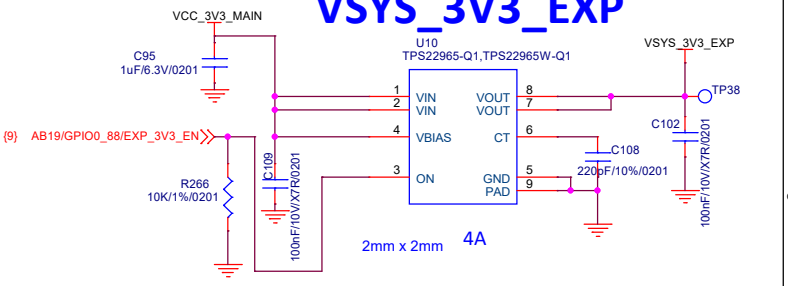
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inhand	Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A			
File:	03. I2C TREE			
Date:	Monday, April 27, 2026		Rev:	V1.0
Designed by:		Reviewed by:	Sheet:	3 of 22





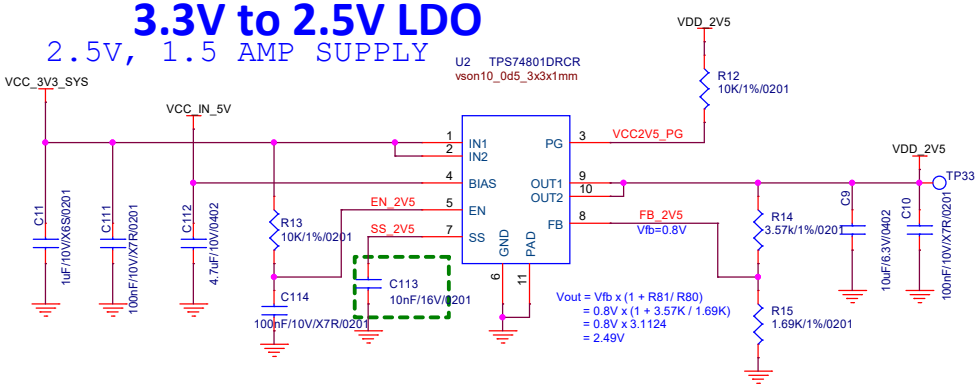
VCC_3V3_MAIN



VSYS_3V3_EXP

3.3V to 2.5V LDO

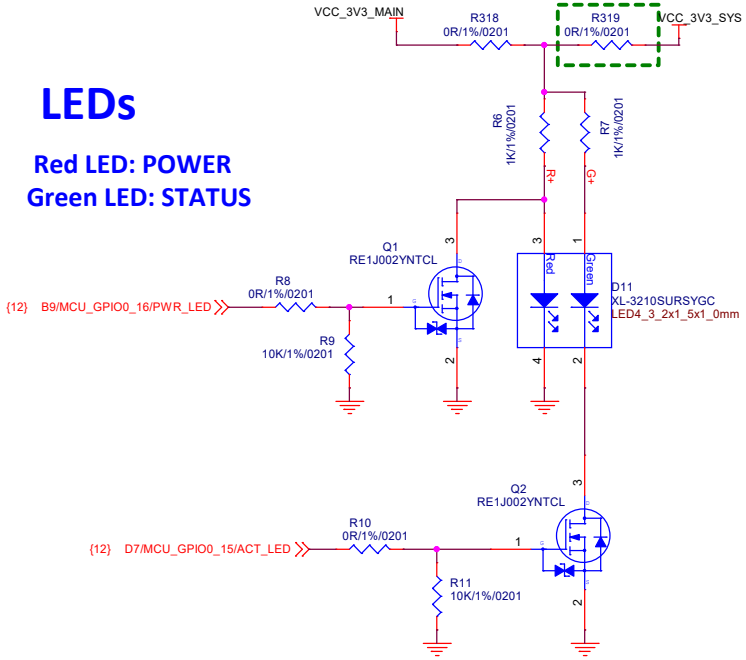
2.5V, 1.5 AMP SUPPLY



Ethernet PHY: VDDA2P5=2.5V, Imax=137mA
LDO U63+u65 Power In=2.5V, Imax=108+80mA=188mA
Total: Imax= 137mA + 188mA = 325mA
Power Consumption: (3.3V-2.5V) * 0.325A = 0.26W

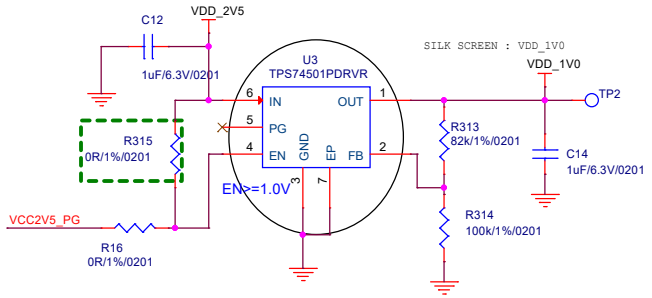
LEDs

Red LED: POWER
Green LED: STATUS



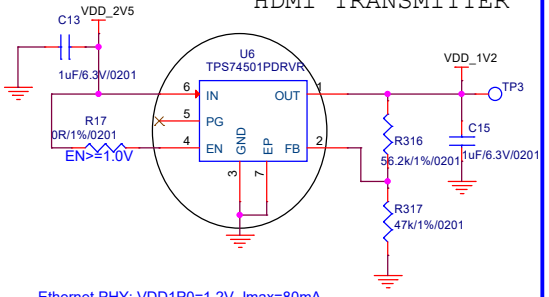
PERIPHERAL SUPPLY - ETHERNET PHY

1.0V, 0.5 AMP SUPPLY



Ethernet PHY: VDD1P0=1.0V, Imax=108mA

1.2V, 0.5 AMP SUPPLY



Ethernet PHY: VDD1P0=1.2V, Imax=80mA

HDMI TRANSMITTER

<Core Design>

Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A		
File:	04. USB TYPE-C & POWER & LED		
Date:	Monday, April 27, 2026	Rev:	V1.0
Designed by:	Reviewed by:	Sheet:	4 of 22

SOC POWER SUPPLY PMIC

PMIC Config option

D-Note :-
Use 10V rated caps to take care of the DC bias and derating

Route as Pseudo differential pair trace

Cad Note :-
Tie the feedback after the DC/DC output bulk caps or use remote sensing

U27 (See "PCB Notes")
TPS65931211RWERQ1

Dual-function PMIC control output, Low Power Mode (active low) or PMIC Enable (active high)

Capacitors C318, C317, C391, C390 & C340 are marked DNP to meet PMIC output cap specifications

Mount R121 & Demount R110 when VCC_CORE is 0.75V

PMIC uses default I2C0 ADDR: 0x48,0x49, 0x5A, 0x5B

R-Note :-
CORE_PWR_STRAP/ GPIO6
High: Buck1/2/3 = 0.85V
LOW/Floating: Buck1/2/3 = 0.75V

VCC_3V3_SYS

VCC_3V3_SYS LOAD SWITCH

TPS22965-Q1,TPS22965W-Q1

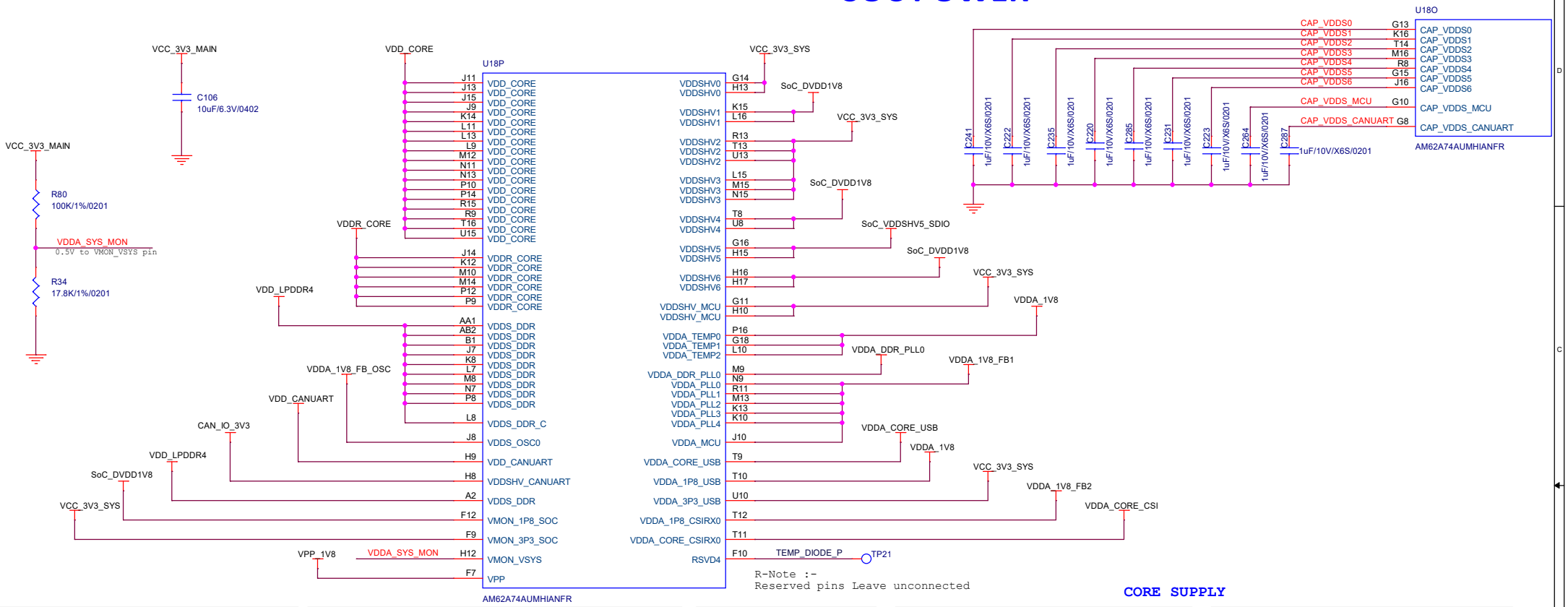
SN74LVC1G08DSF

260423:V1.1

Project:	Mo_62A
File:	05. PMIC
Date:	Monday, April 27, 2026
Rev:	V1.0
Designed by:	
Reviewed by:	
Sheet:	5 of 22

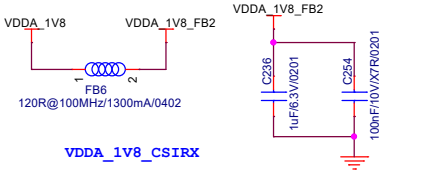
Beijing InHand Networks Technology Co., Ltd.

SOC POWER

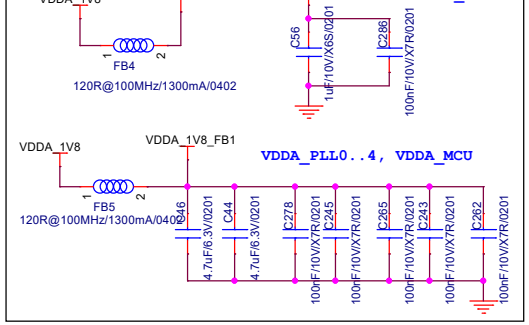


CORE SUPPLY

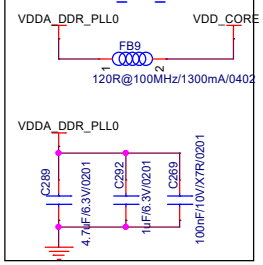
1.8V Analog SUPPLY



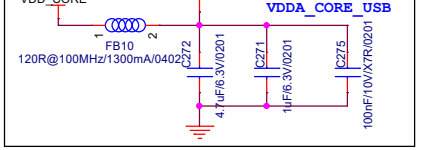
VDDS_OSC0



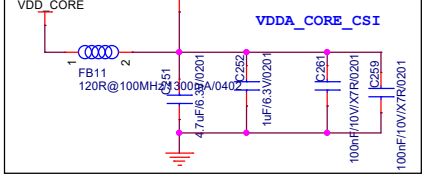
VDDA_DDR_PLL0



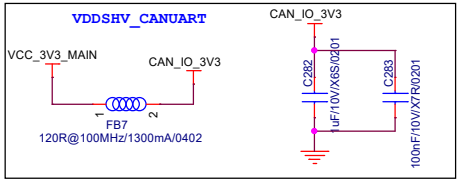
VDDA_CORE_USB



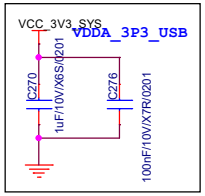
VDDA_CORE_CSI



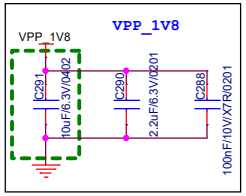
VDDSHV_CANUART



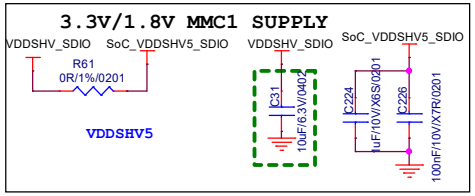
VDDA_3P3_USB



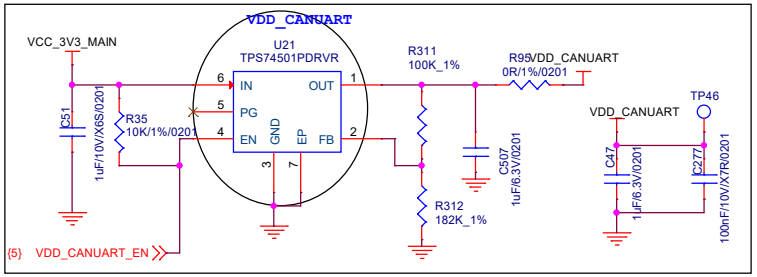
VPP_1V8



3.3V/1.8V MMC1 SUPPLY



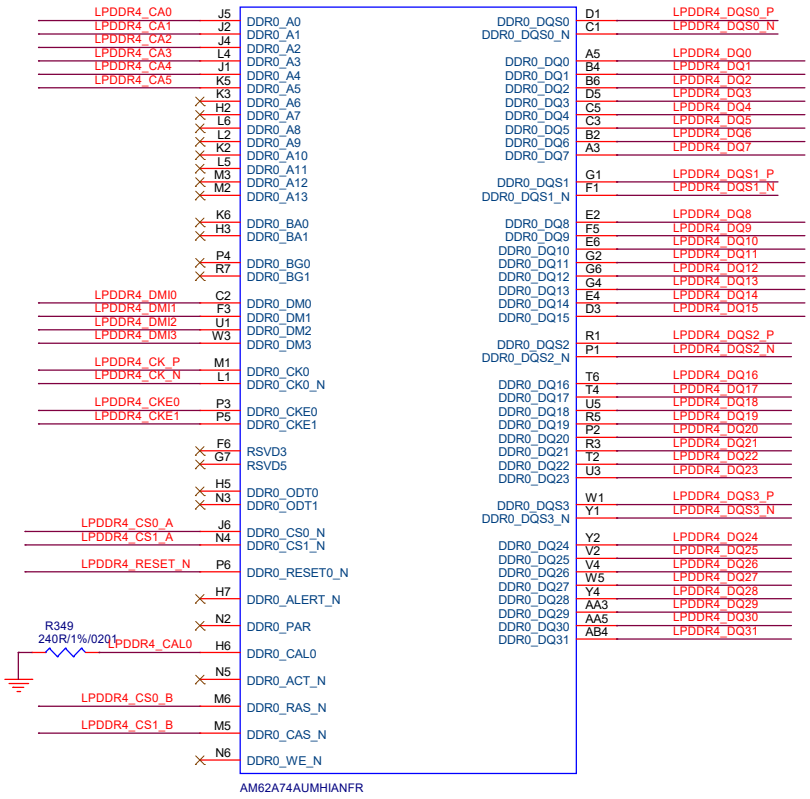
VDD_CANUART

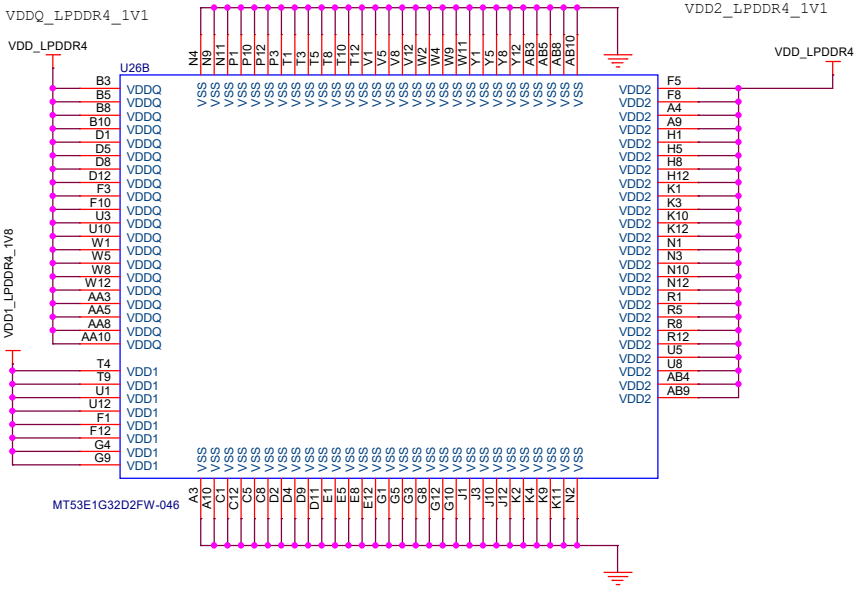


<Core Design>

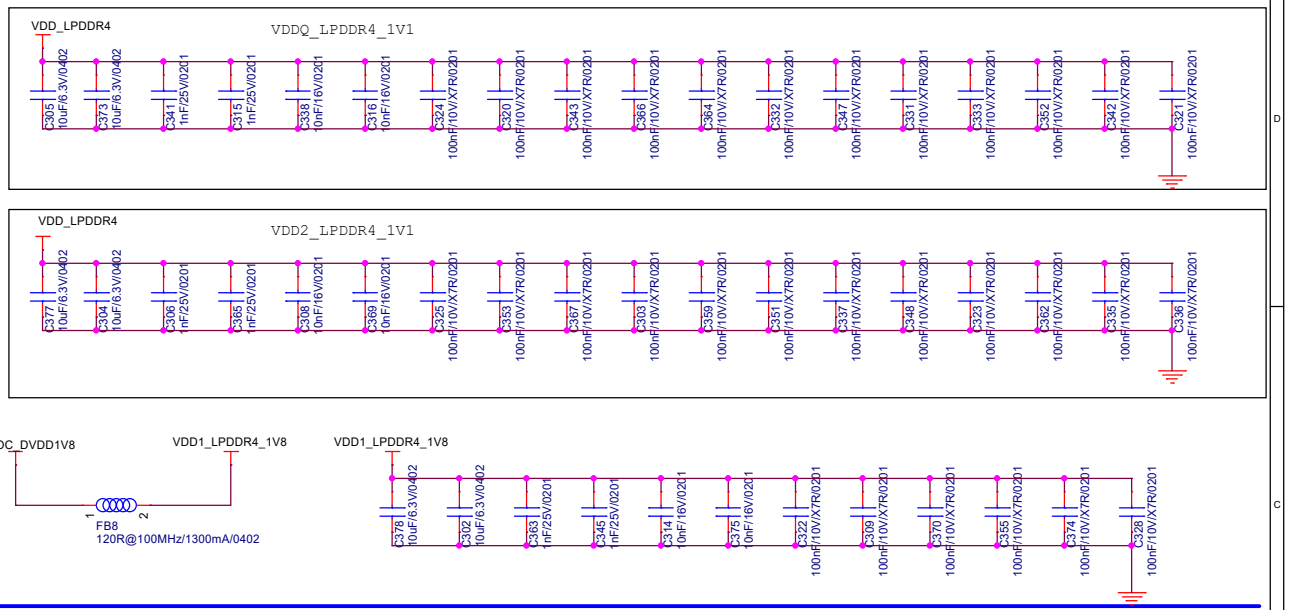
inhand	Beijing InHand Networks Technology Co., Ltd.		
Project:	Mo_62A		
File:	06. SOC POWER		
Date:	Monday, April 27, 2026	Rev:	V1.0
Designed by:	Reviewed by:	Sheet:	6 of 22

SOC LPDDR4 INTERFACE

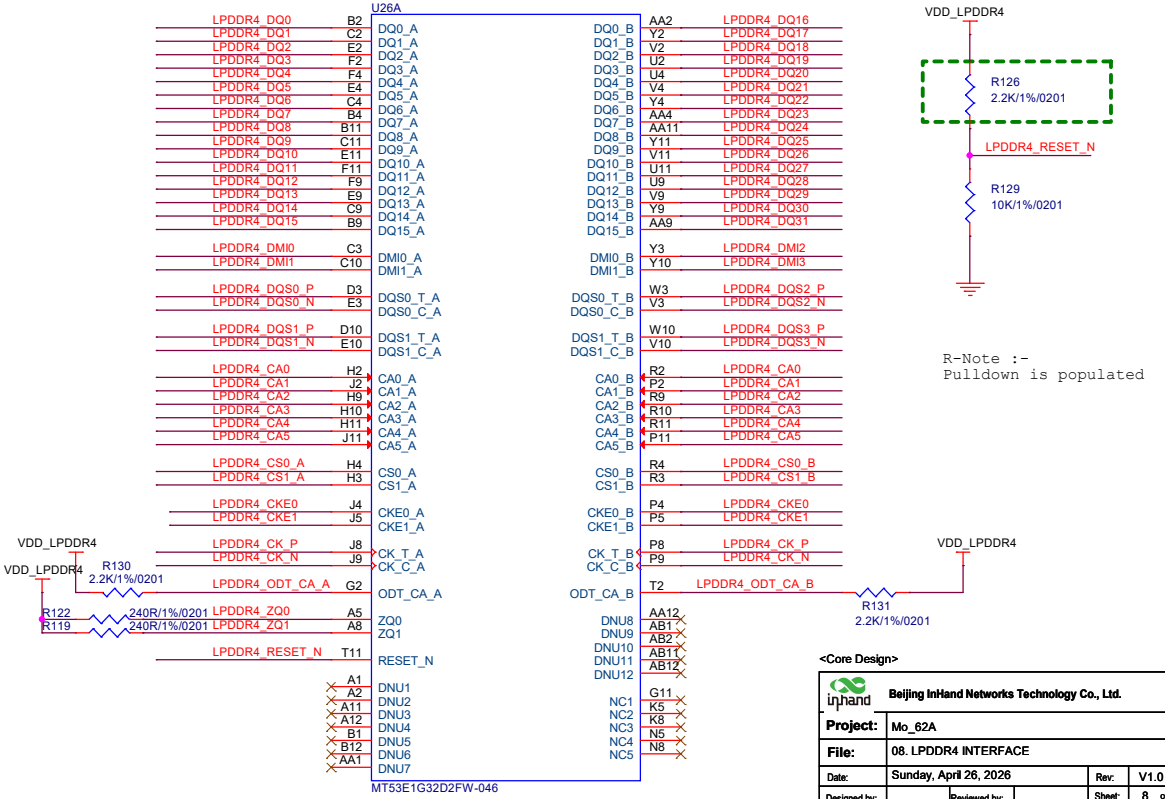




LPDDR4 POWER DECAPS



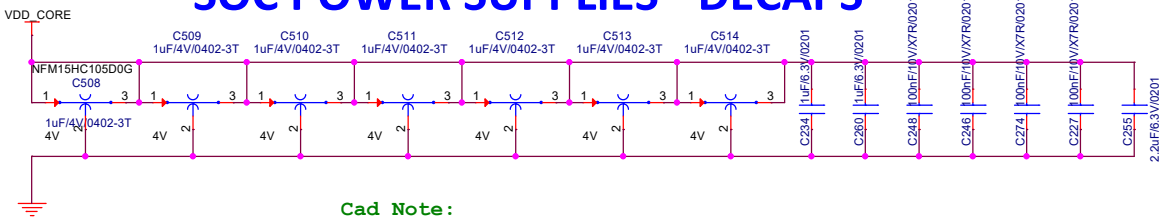
LPDDR4 DEVICE



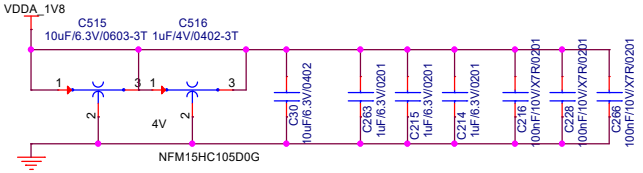
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Beijing InHand Networks Technology Co., Ltd.	
Project:	Mo_62A
File:	08. LPDDR4 INTERFACE
Date:	Sunday, April 26, 2026
Rev:	V1.0
Designed by:	
Reviewed by:	
Sheet:	8 of 22

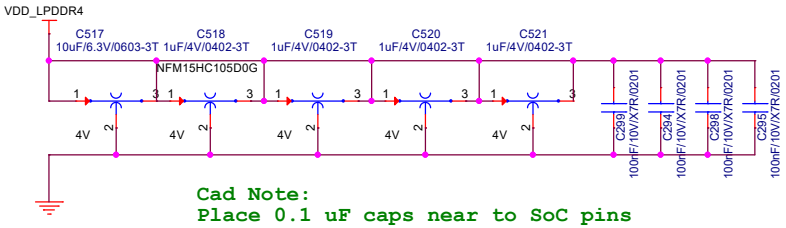
SOC POWER SUPPLIES - DECAPS



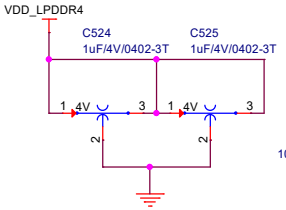
Cad Note:
Place 0.1 uF caps near to SoC pins

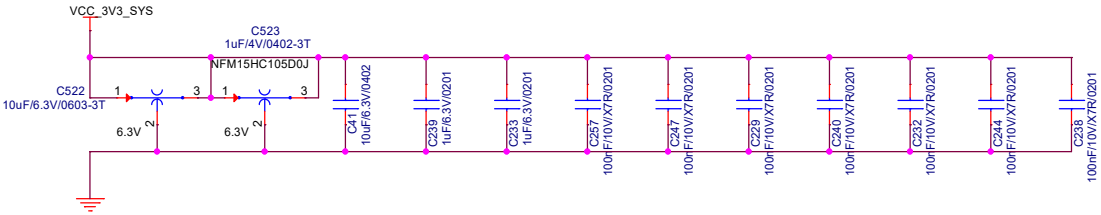


Cad Note:
Place 0.1 uF caps near to SoC pins

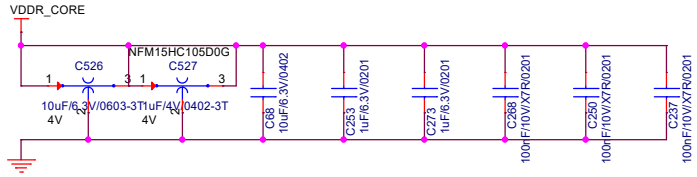


Cad Note:
Place 0.1 uF caps near to SoC pins



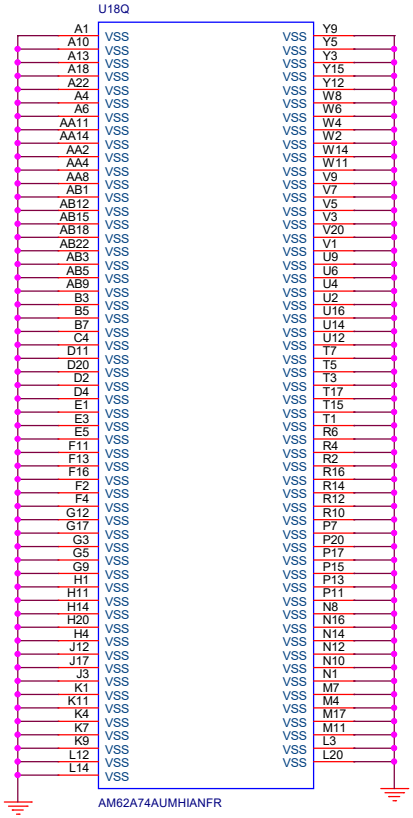


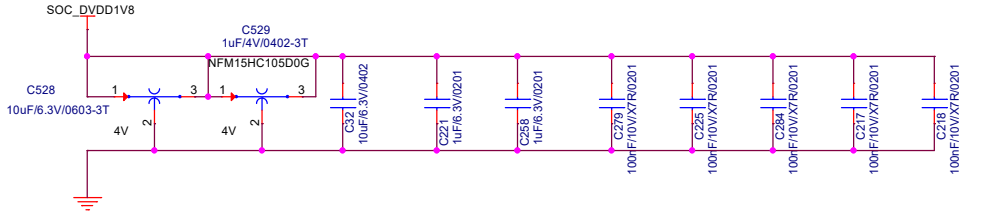
Cad Note:
Place 0.1 uF caps near to SoC pins



Cad Note:
Place 0.1 uF caps near to SoC pins

SOC VSS



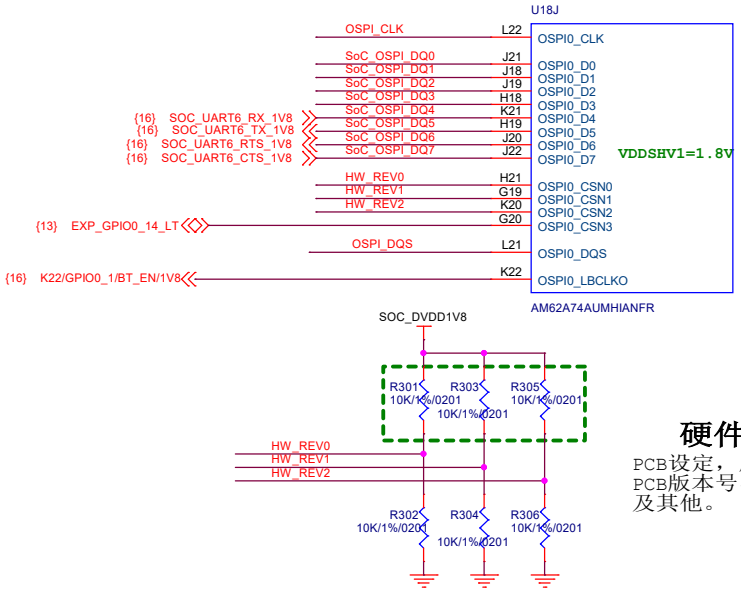


Cad Note:
Place 0.1 uF caps near to SoC pins

<Core Design>

inhand	Beijing InHand Networks Technology Co., Ltd.		
Project:	Mo_62A		
File:	07. SOC POWER CAPS		
Date:	Sunday, April 26, 2026	Rev:	V1.0
Designed by:	Reviewed by:	Sheet:	7 of 22

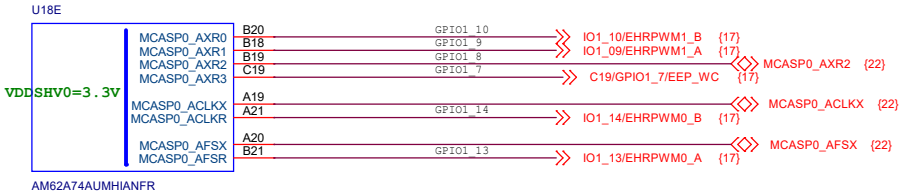
SOC OSPI INTERFACE



硬件设置

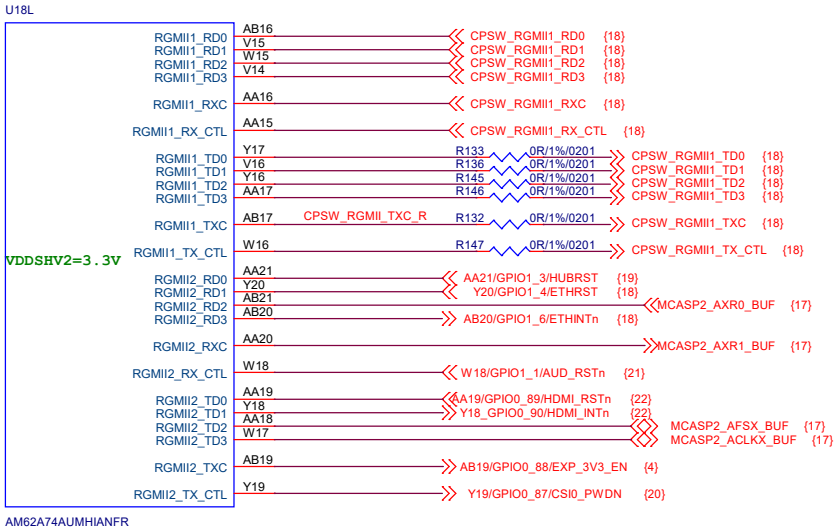
PCB设定，用于设置PCB版本号、DDR型号、及其他。

SOC MCASP0 INTERFACE

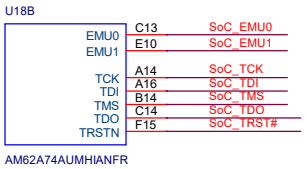


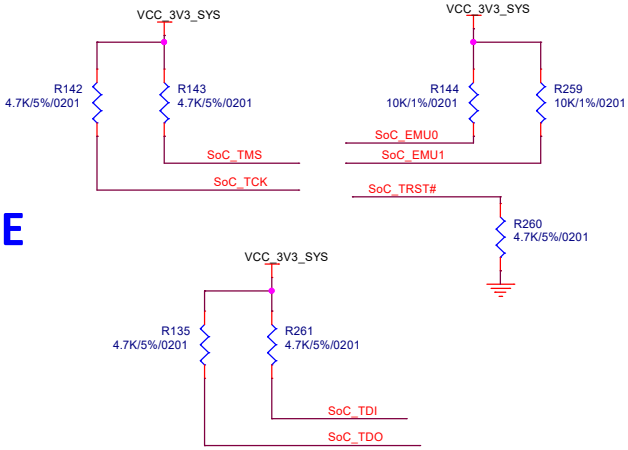
GPIO1_13-----SOC_SPI2_CS0
GPIO1_14-----SOC_SPI2_CLK
GPIO1_7 -----SOC_SPI2_D0
GPIO1_8 -----SOC_SPI2_D1

SOC MAC INTERFACE



SOC JTAG INTERFACE

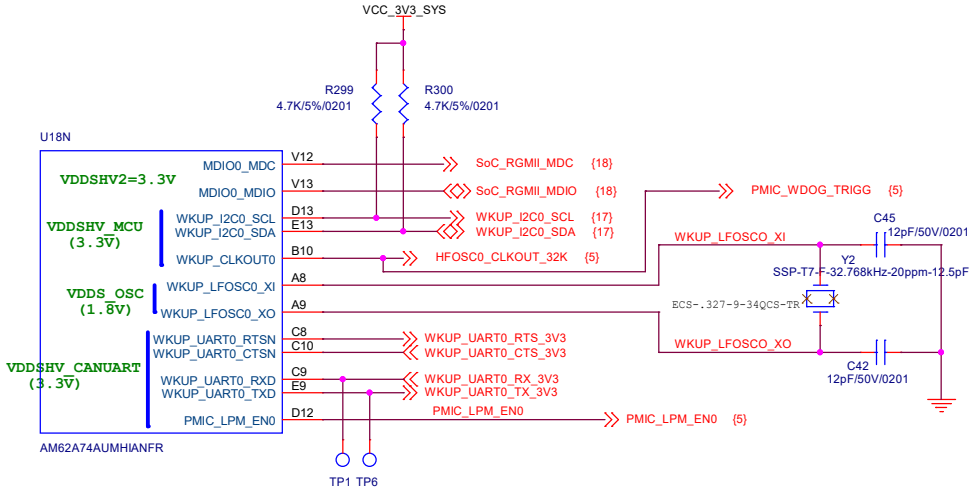




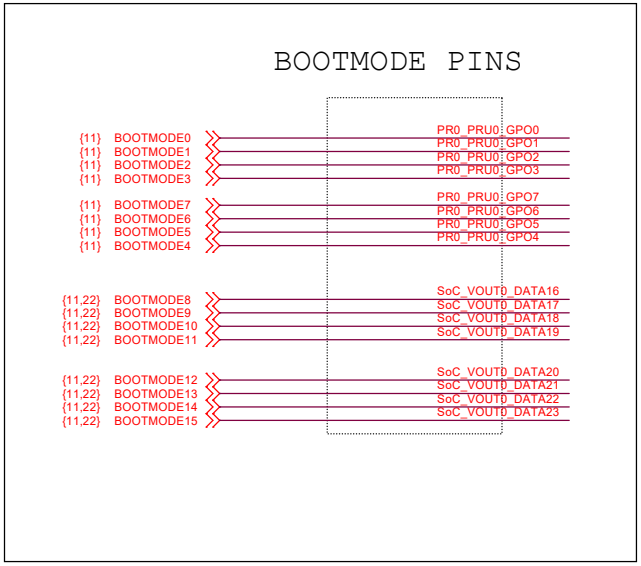
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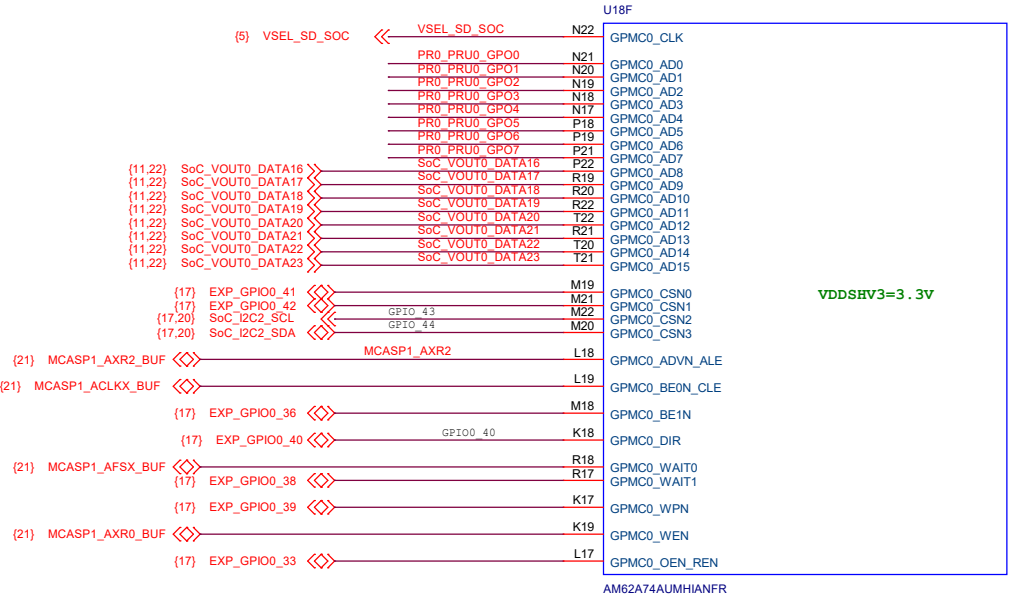
inhand	Beijing InHand Networks Technology Co., Ltd.		
Project:	Mo_62A		
File:	09. SOC MAC,MCASP0,OSPI,JTAG		
Date:	Sunday, April 26, 2026	Rev:	V1.0
Designed by:	Reviewed by:	Sheet:	9 of 22

SOC WKUP DOMAIN



SOC GPMC INTERFACE





<Core Design>

inhand	Beijing InHand Networks Technology Co., Ltd.		
Project:	Mo_62A		
File:	10. SOC WKUP & GPMC		
Date:	Sunday, April 26, 2026	Rev:	V1.0
Designed by:	Reviewed by:	Sheet:	10 of 22

BOOTMODE CONFIGURATION RESISTORS

BOOTMODE7 {10}
BOOTMODE6 {10}
BOOTMODE5 {10}
BOOTMODE4 {10}
BOOTMODE3 {10}
BOOTMODE2 {10}
BOOTMODE1 {10}
BOOTMODE0 {10}

BOOTMODE15 {10,22}
BOOTMODE14 {10,22}
BOOTMODE13 {10,22}
BOOTMODE12 {10,22}
BOOTMODE11 {10,22}
BOOTMODE10 {10,22}
BOOTMODE9 {10,22}
BOOTMODE8 {10,22}

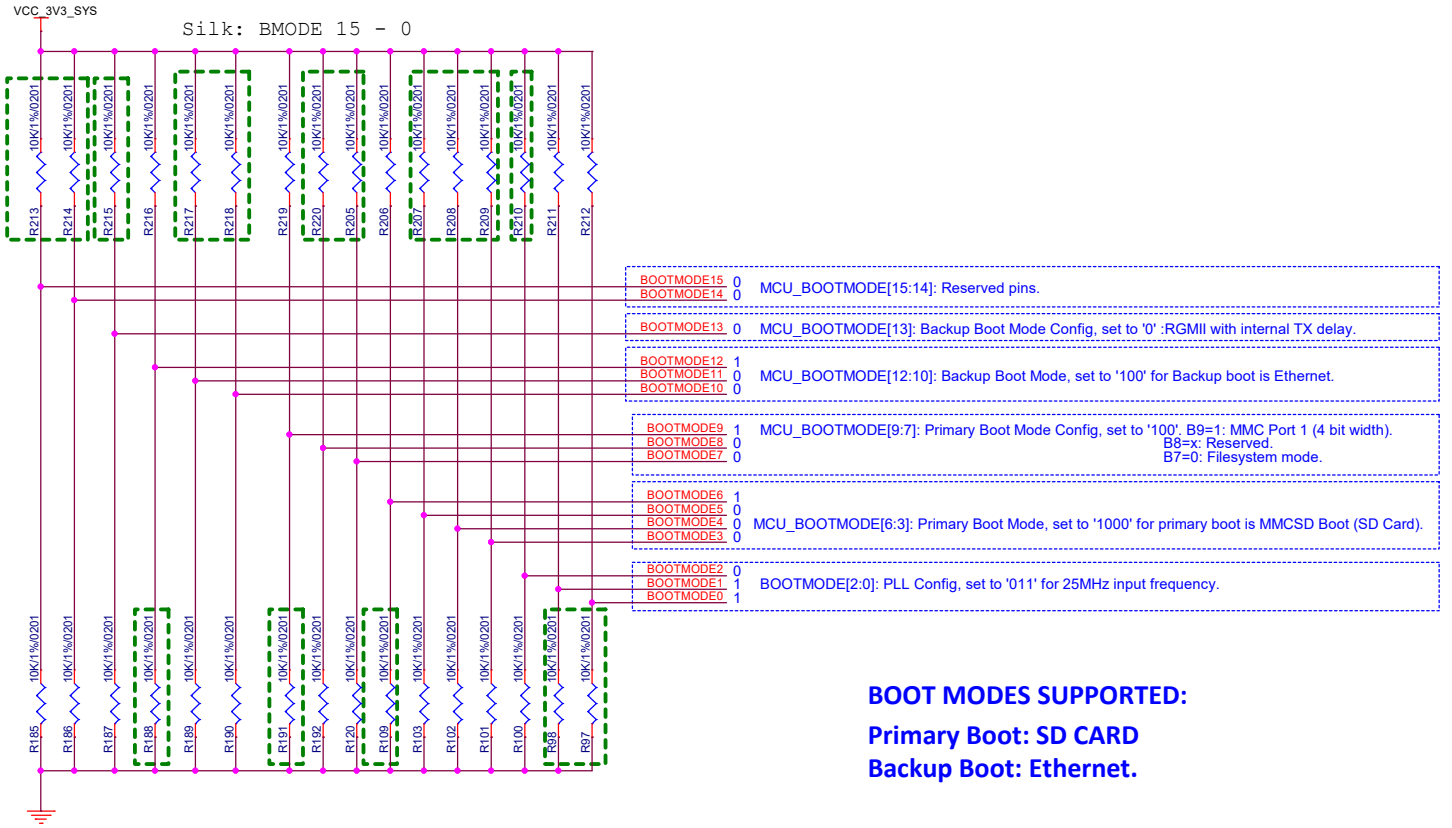
D-Note :-
VCC3V3_XDS_TA supply is used to support test automation.
Connect SOC_DVDD3V3 in the custom board design when buffers are not used

D-Note :-
it is not recommended to leave any of the the bootmode pin open including reserved pins

D-Note :-
Dip switch is optional and used on the SK for ease of configuration
A pullup or pulldown resistor can be used to set the BOOTMODE configuration
Provide provision for Pullup and Pulldown resistors for the bootmode pins that have configuration capability

D-Note :-
When dip switches are used on custom board, an external ESD protection may be required if the DIP switches are expected to be configured in an uncontrolled ESD environment

D-Note :-
When DIP switch are used, reduce the resistor values used for the divider to 47K and 470R maintaining the ratio



BOOT MODES SUPPORTED:

Primary Boot: SD CARD

Backup Boot: Ethernet.

BOOT MODES SUPPORTED

1. OSPI
2. MMC1 - SD CARD
3. UART
4. eMMC
5. ETHERNET
6. USB0 DFU
7. USB0 MS

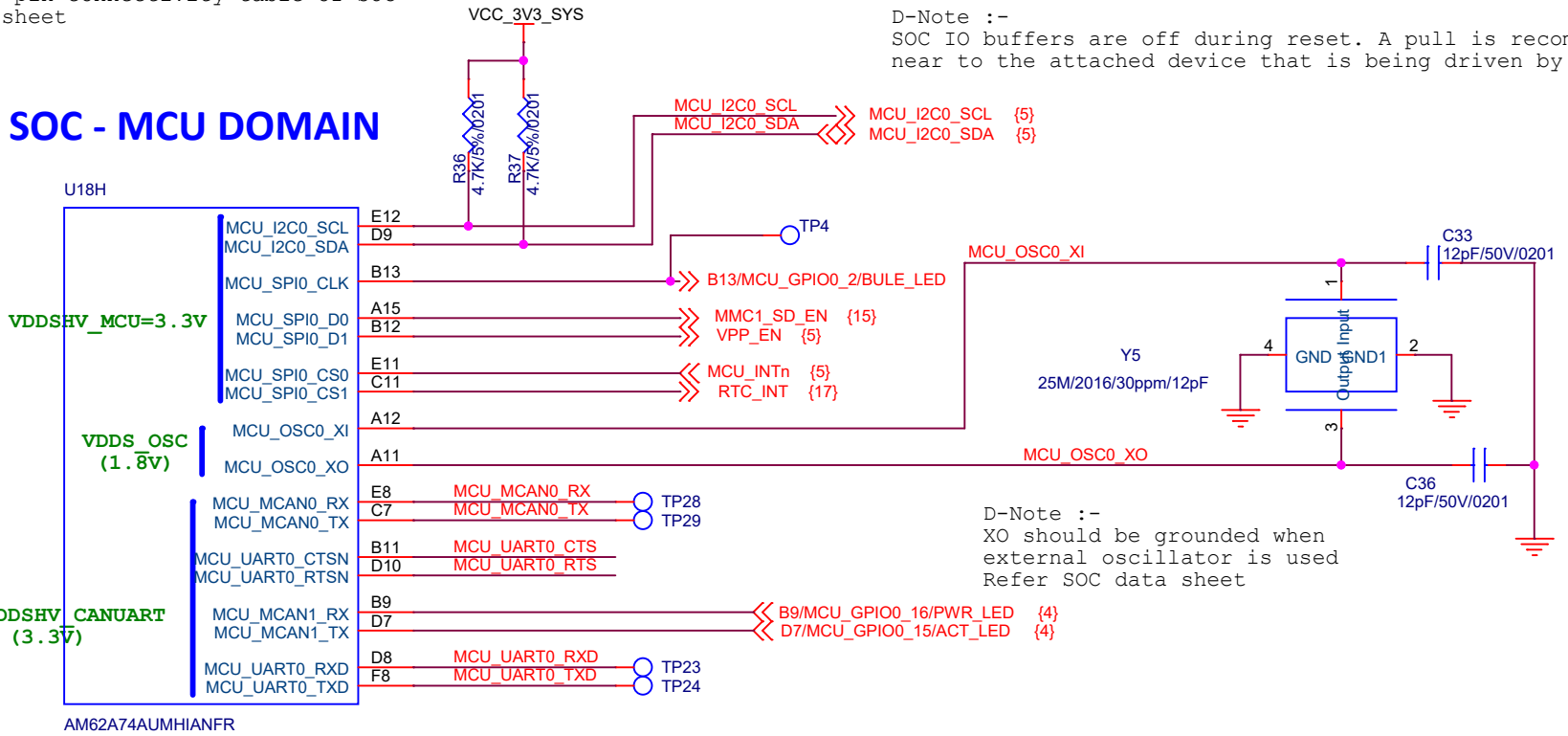
<Core Design>

inhand	Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A			
File:	11. BOOT MODE			
Date:	Sunday, April 26, 2026		Rev:	V1.0
Designed by:		Reviewed by:	Sheet:	11 of 22

D-Note :-
A pullup is recommended for Open-drain output type I2C interfaces irrespective of the IO configuration
Refer pin connectivity table of SOC data sheet

D-Note :-
Open-drain output type buffer I2C interfaces have slew rate requirement when pulled to 3.3 V An RC is recommended for slew rate control.
Refer SK-AM62P-LP schematics

D-Note :-
SOC IO buffers are off during reset. A pull is recommended near to the attached device that is being driven by the SOC IO



D-Note :-
XO should be grounded when external oscillator is used
Refer SOC data sheet

D-Note :-
Connect the 25 MHz crystal directly to the SOC Xi and Xo pins (No Series or parallel resistors are recommended). The internal oscillator implements AGC (Automatic Gain Control) for amplitude control. Match the SOC and the EPHY crystal specs

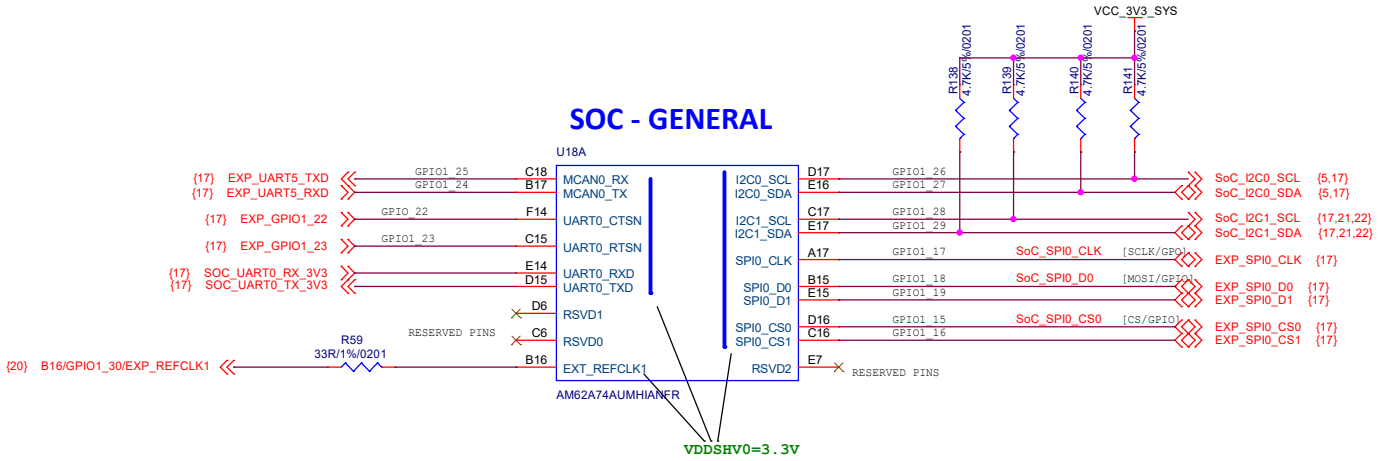
D-Note :-
Refer Applications, Implementation, and Layout section of the data sheet for clock routing guidelines as below:
Clock Routing Guidelines
Oscillator Routing

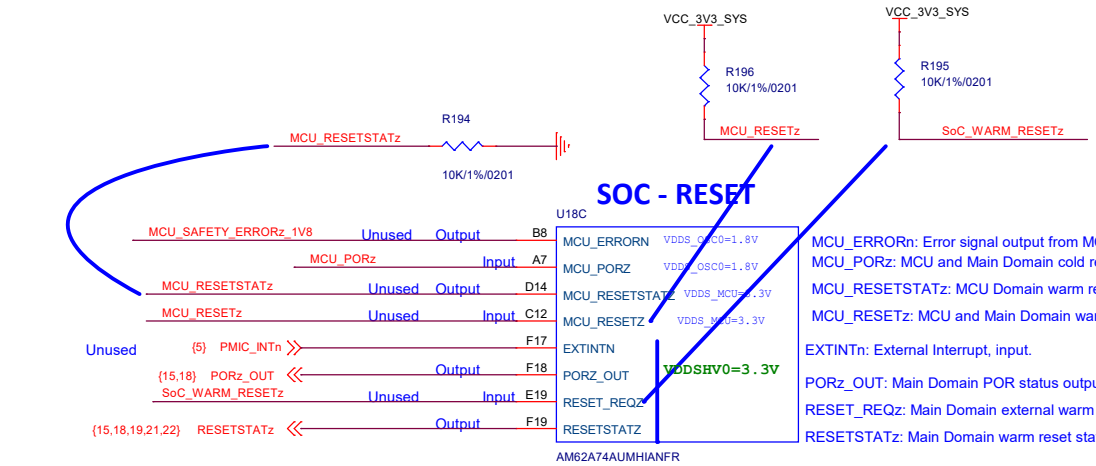
D-Note :-
No HFOSC0 registers are required to be changed. These registers should remain in their default state.
Select the appropriate crystal circuit components that are compliant to the values defined in the MCU_OSC0 Crystal Circuit Requirements table. Read the Load Capacitance and Shunt Capacitance sections to select the appropriate crystal circuit components.

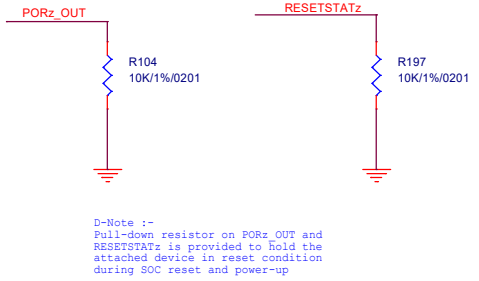
D-Note :-
MCU_OSC0 has been validated only with a 25 MHz clock source, so that is the only frequency supported. The datasheet shows MCU_OSC0 not starting until after the core voltage because there are some cases where the oscillator may not start until VDD_CORE is valid. In most cases it will start as early as VDDS_OSC0, but this may not always be the case. This diagram in the datasheet is showing the maximum start-up time, which must include the case where the delay is based on VDD_CORE being valid.

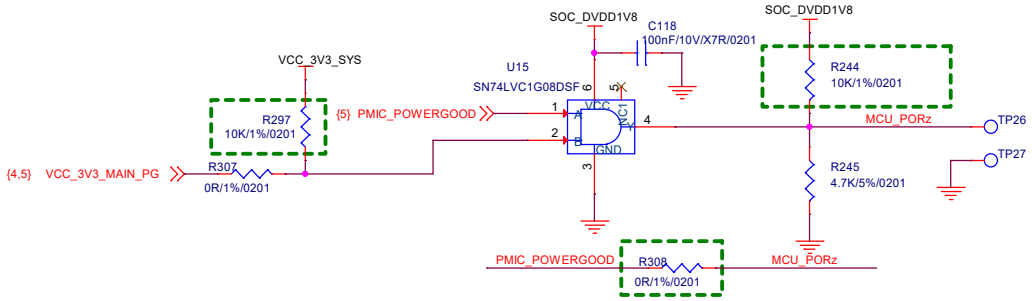
<Core Design>

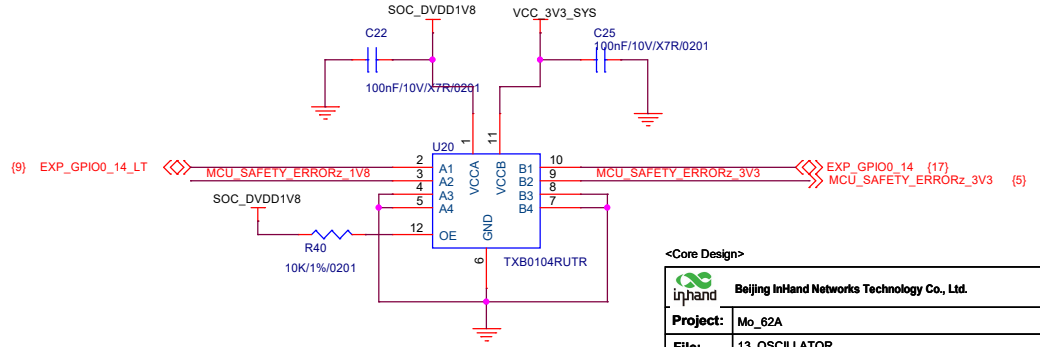
Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A		
File:	12. MCU HEADER		
Date:	Sunday, April 26, 2026	Rev:	V1.0
Designed by:	Reviewed by:	Sheet:	12 of 22







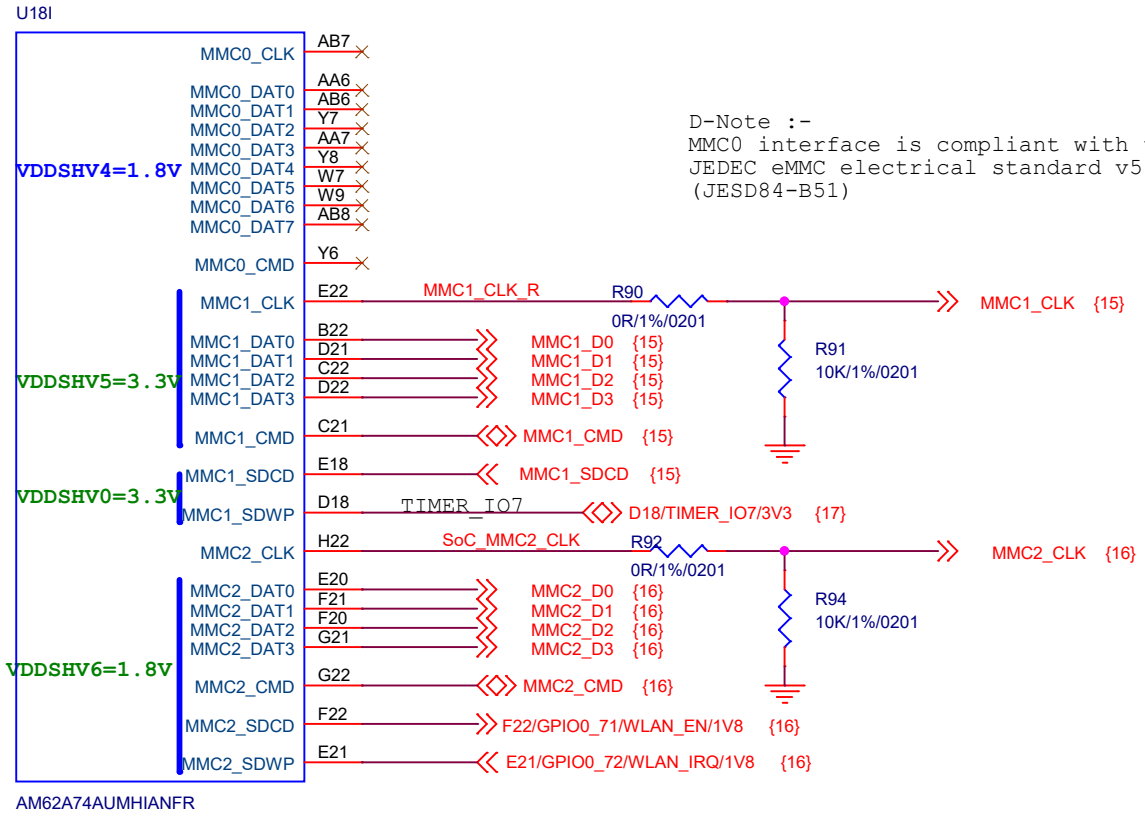




<Core Design>

inhand	Beijing InHand Networks Technology Co., Ltd.				
Project:	Mo_62A				
File:	13. OSCILLATOR				
Date:	Sunday, April 26, 2026		Rev:	V1.0	
Designed by:		Reviewed by:		Sheet:	13 of 22

SOC - MMC Interface



D-Note :-
MMC0 interface is compliant with the JEDEC eMMC electrical standard v5.1 (JESD84-B51)

<Core Design>

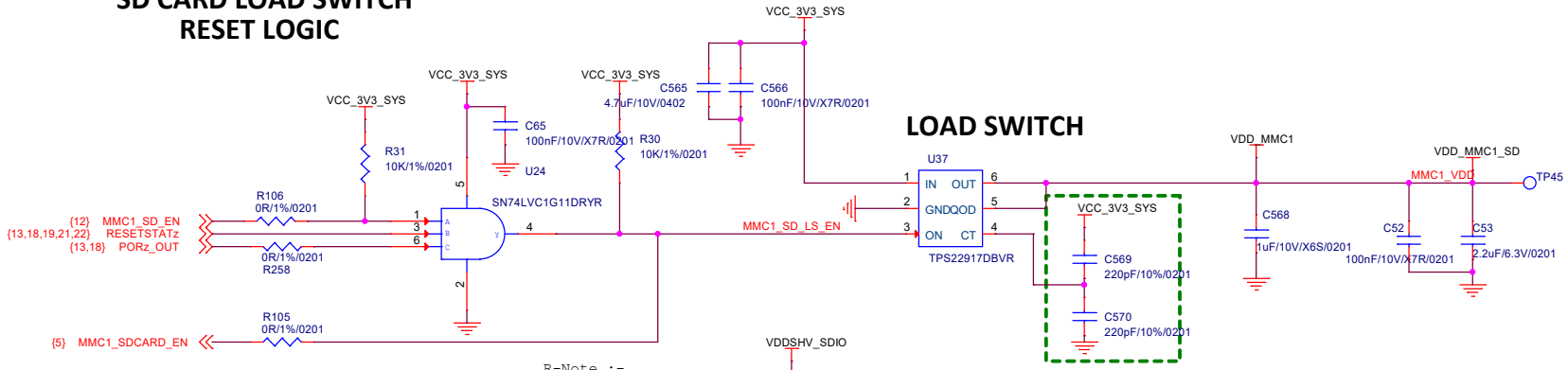
inhand	Beijing InHand Networks Technology Co., Ltd.		
Project:	Mo_62A		
File:	14. MMCx		
Date:	Sunday, April 26, 2026	Rev:	V1.0
Designed by:	Reviewed by:	Sheet:	14 of 22

D-Note :-
This power switch, along with the reset logic, and the host IO power supply circuit is required to support UHS-I SD Cards which begins communications using 3.3V signal levels and later change to 1.8V signal levels when changing to one of the faster data transfer speeds. Cycling power to the SD Card is the only way to put it back into 3.3V mode since SD Cards do not have a reset pin. The host IO power supply must power off/on and change voltage at the same time as the SD Card. These circuits and the software driver operating the signals sourcing these circuits ensure both devices are off, or on and operating at the same IO voltage at the same time.

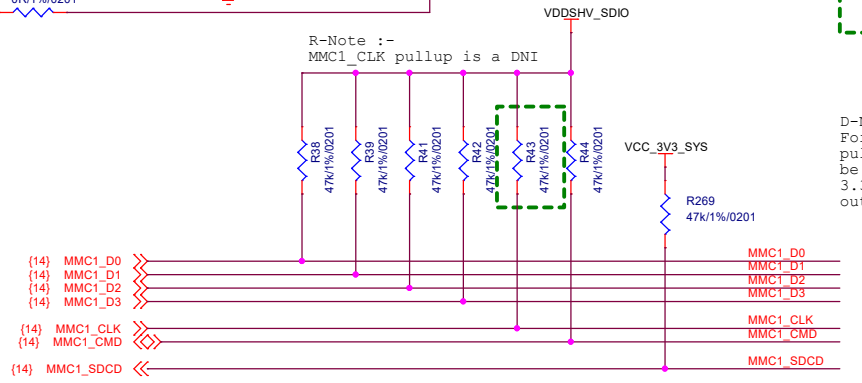
SD CARD INTERFACE

SD CARD LOAD SWITCH RESET LOGIC

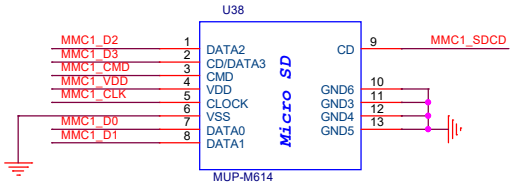
LOAD SWITCH

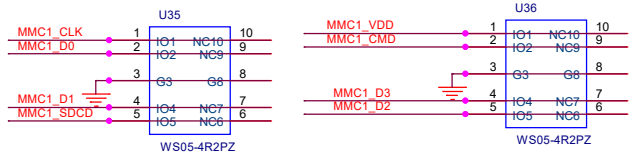


R-Note :-
MMC1_CLK pullup is a DNI



D-Note :-
For UHS-I operation, the pullups are recommended to be connected to the 3.3V/1.8V switched LDO output

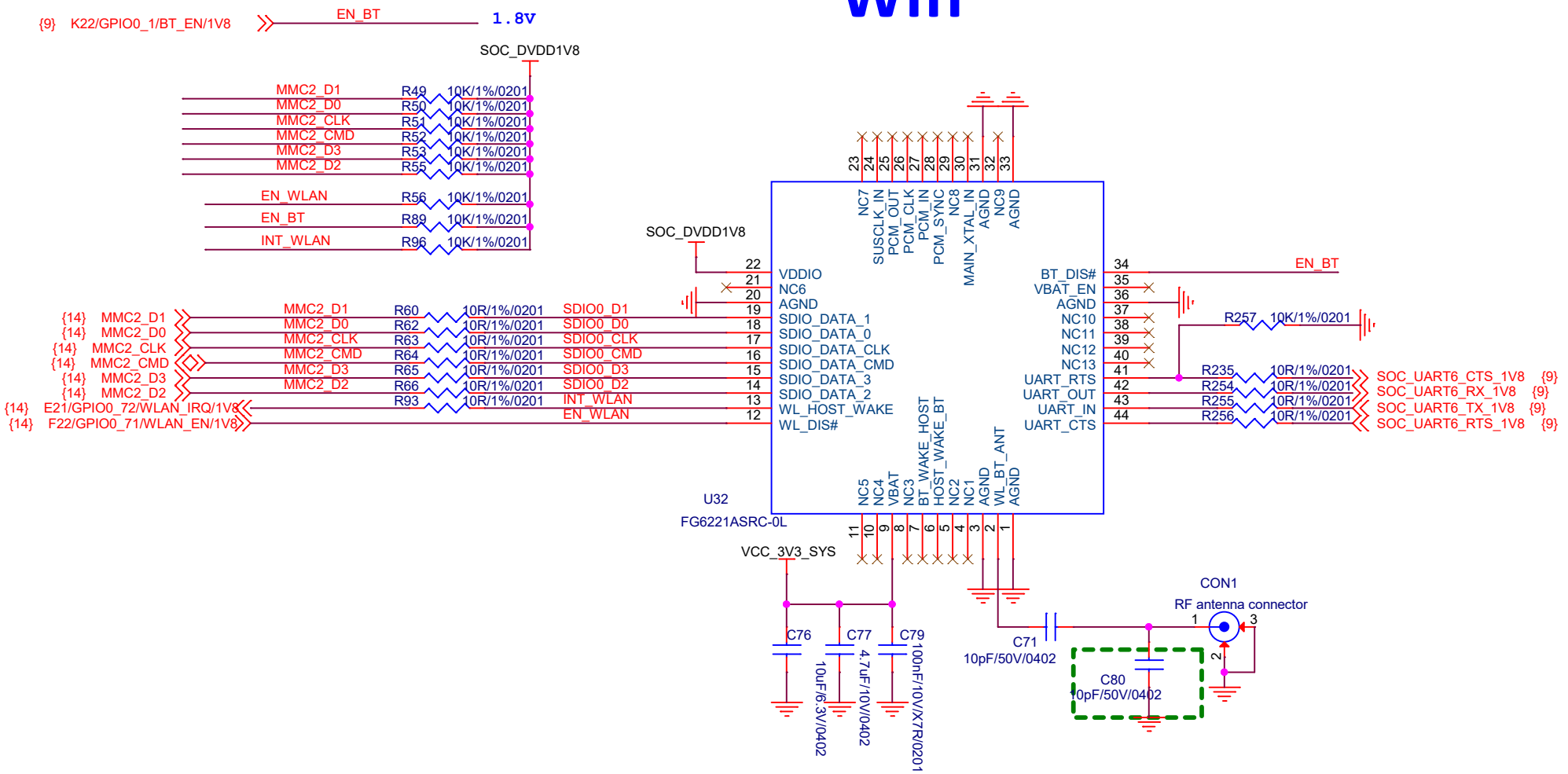




<Core Design>

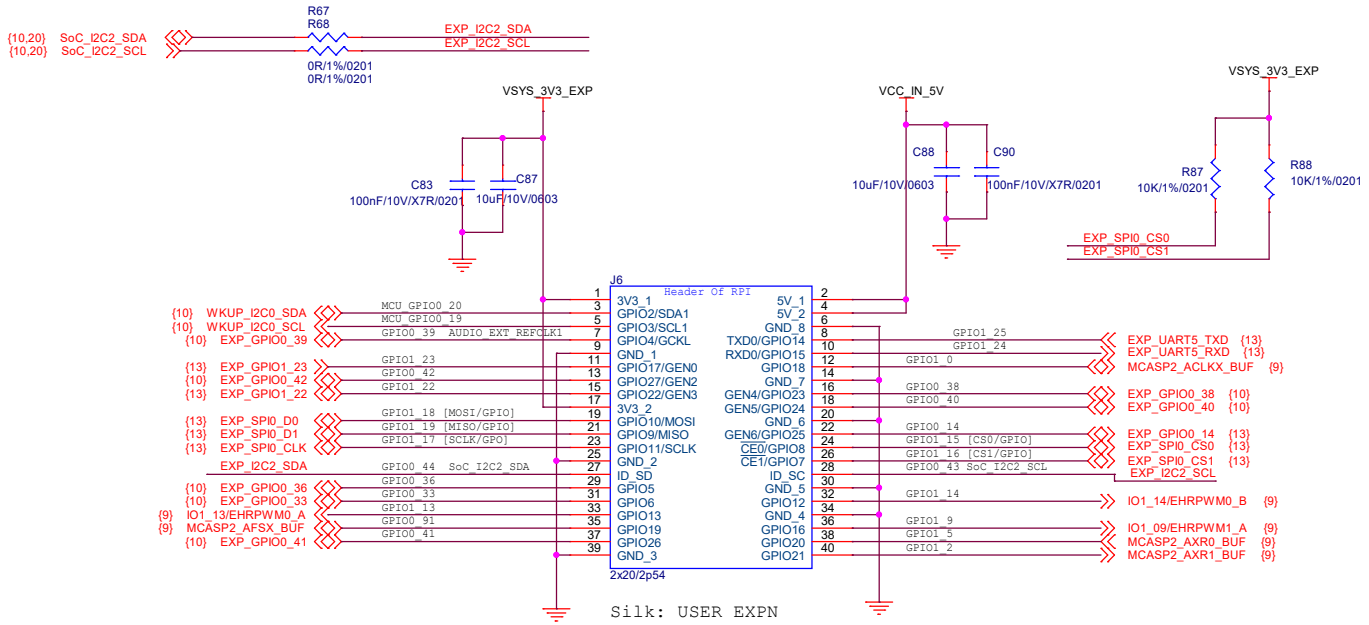
inhand	Beijing InHand Networks Technology Co., Ltd.			
Project:	Mo_62A			
File:	15. SD CARD INTERFACE			
Date:	Sunday, April 26, 2026		Rev:	V1.0
Designed by:		Reviewed by:	Sheet:	15 of 22

Wifi

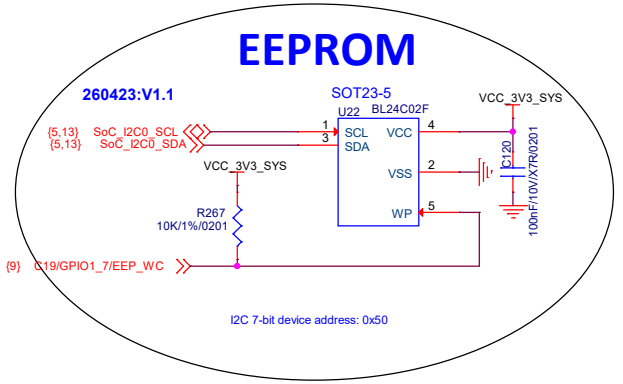


<Core Design>

inhand	Beijing InHand Networks Technology Co., Ltd.				
Project:	Mo_62A				
File:	16. WIFI/BLE				
Date:	Sunday, April 26, 2026			Rev:	V1.0
Designed by:		Reviewed by:		Sheet:	16 of 22

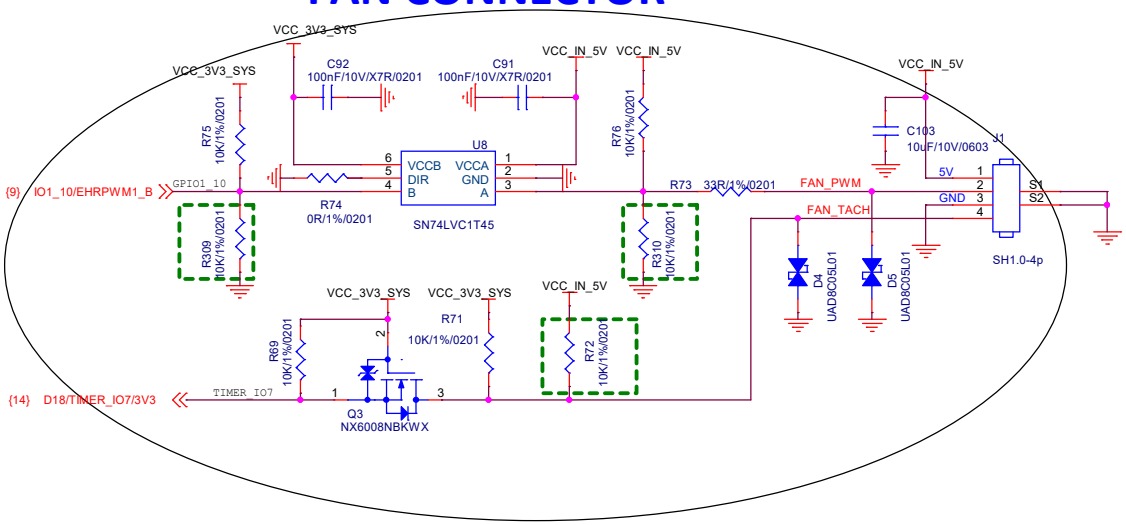


EEPROM

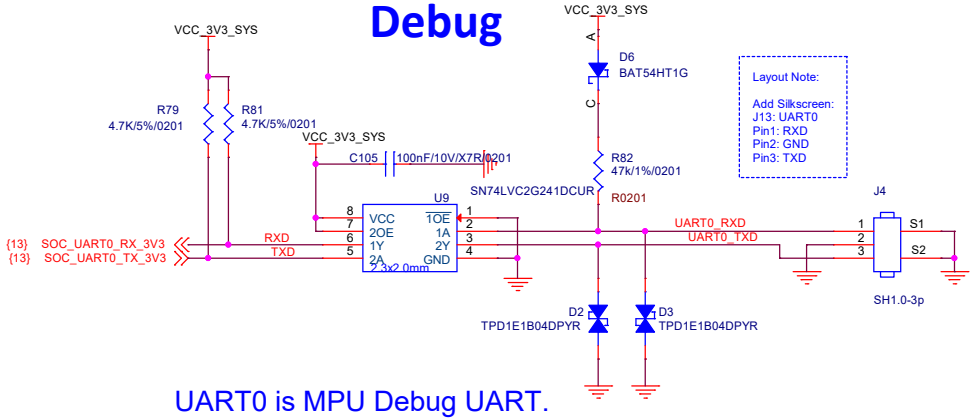


FAN CONNECTOR

260423:V1.1

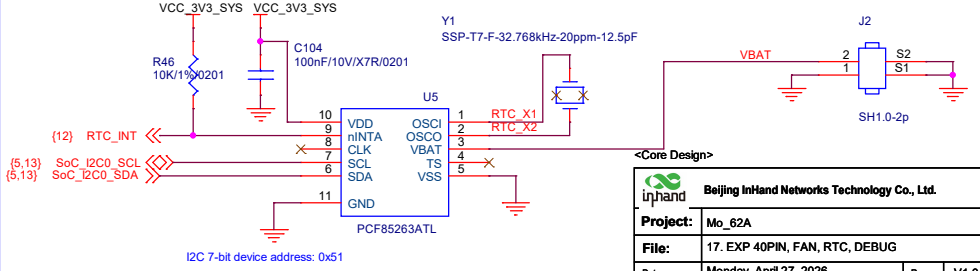


Debug



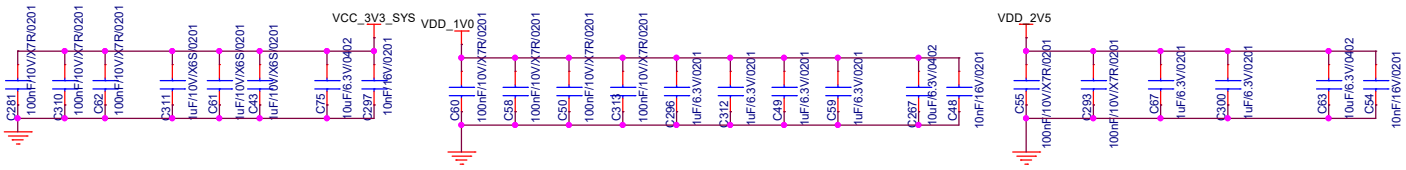
UART0 is MPU Debug UART.

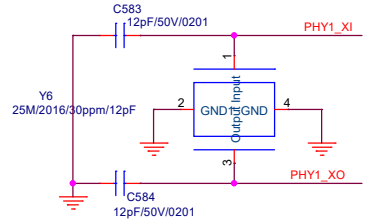
IIC EXT RTC

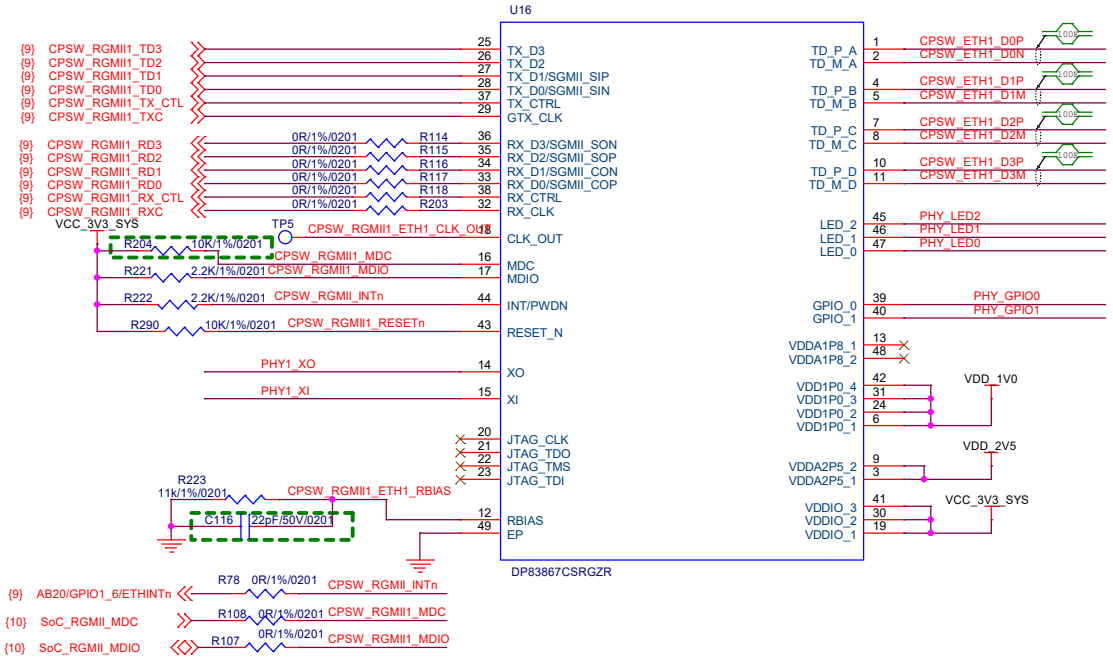


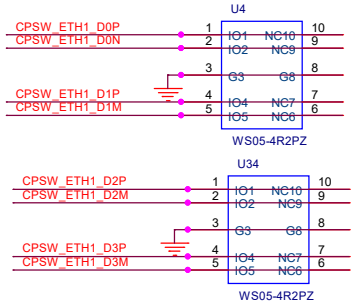
inhand	Beijing InHand Networks Technology Co., Ltd.		
Project:	Mo_62A		
File:	17. EXP 40PIN, FAN, RTC, DEBUG		
Date:	Monday, April 27, 2026	Rev:	V1.0
Designed by:	Reviewed by:	Sheet:	17 of 22

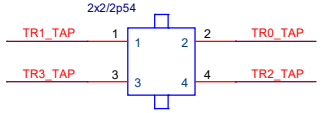
CPSW3G RGMII 1 - ETHERNET PHY





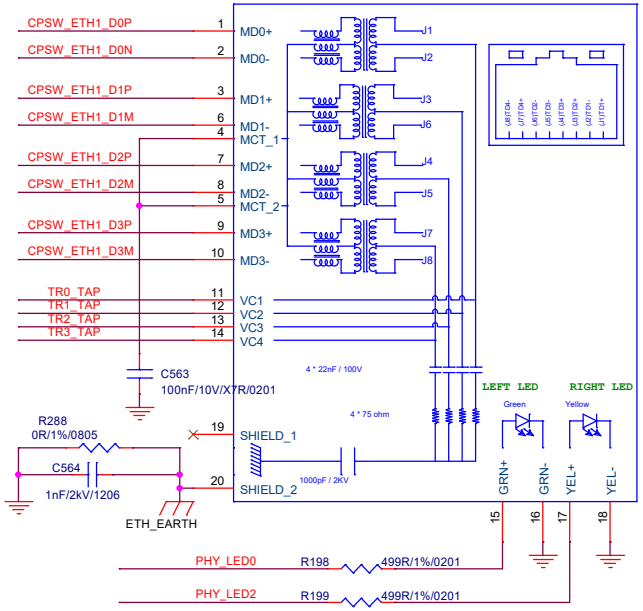






PoE HEADER

RJ45 CONNECTOR WITH INTEGRATED MAGNETICS



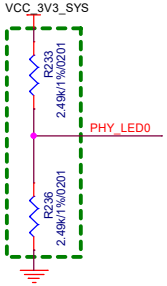
LED0: Link
LED1: Link_1000BT(Unused)
LED2: ACT

```
PHY ADDRESS = 00000
Auto-negotiation Enabled
10/100/1000 advertised, Auto-MDI-X
Tx Clock Skew = 0ns
Rx Clock Skew = 2ns
```

MODE	Rhi	Rlo
1	NC	NC
2	10k	2.49k
3	5.76k	2.49k
4	2.49k	NC

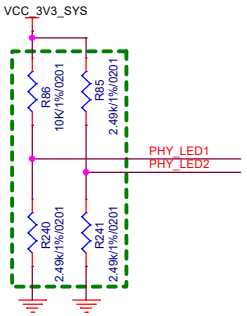
PHY address RXD2、RXD0 set to 0X00

RD2: MODE 1 & RD0: MODE 1
PHY ADD[3:0]=[0000]
PHY ADDRESS = 00000

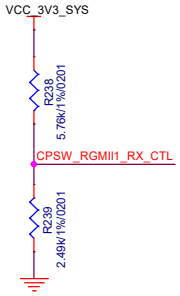


LED_0: MODE 1
Mirror Enable=[0]
For Enabling Mount R233, R236

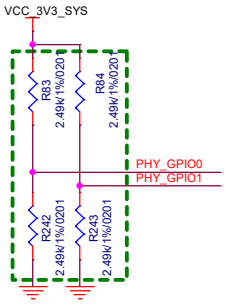
LED1 & LED2: MODE1
RGMII Clock Skew TX[0:2]=[000]
ANEG_SEL(Auto-negotiation)=[0] mean 10/100/1000
PHY_LED[0:2] CANT CONFIGURE IN MODE4



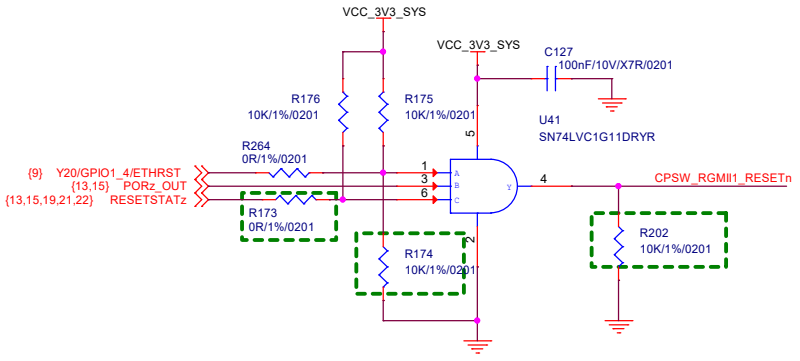
LED1: MODE2 & LED2: MODE1
RGMII Clock Skew TX[2:0]=[000]
ANEG_SEL(Auto-negotiation)=[0]



RX_CTRL:MODE 3
Autoneg enable.
*For Disabling R238=2.49K, R239=NI



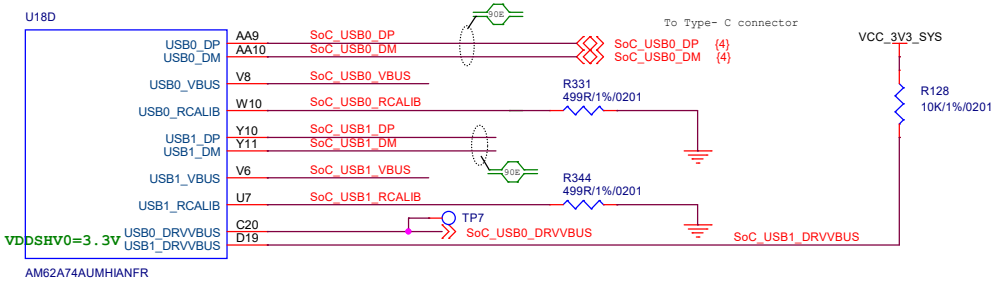
GPIO1 & GPIO0=MODE 1
RGMII Clock Skew RX[0:2]=[000]

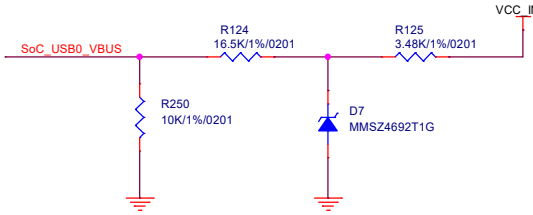


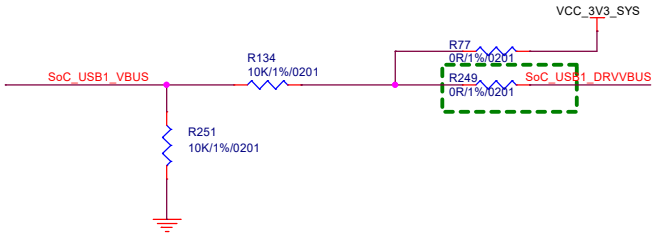
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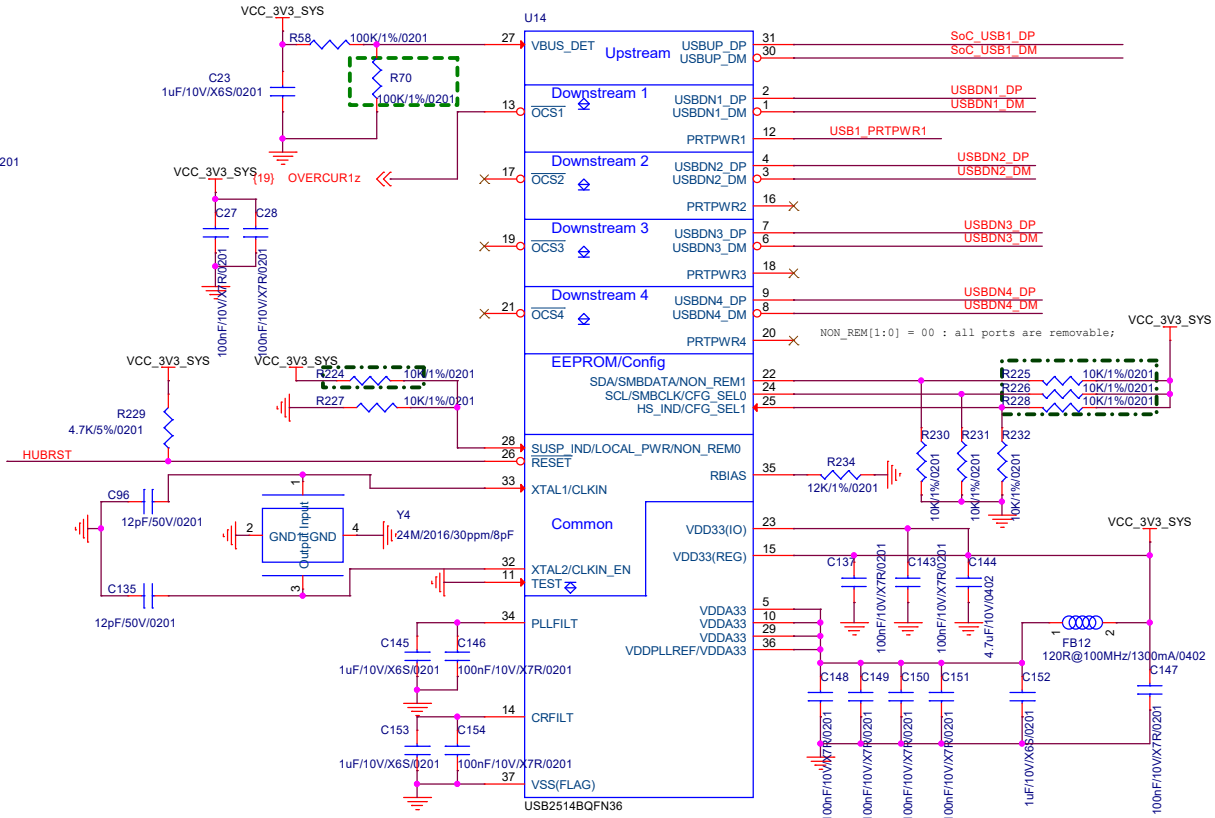
inhand	Beijing InHand Networks Technology Co., Ltd.		
Project:	Mo_62A		
File:	18. ETHERNET		
Date:	Sunday, April 26, 2026	Rev:	V1.0
Designed by:	Reviewed by:	Sheet:	18 of 22

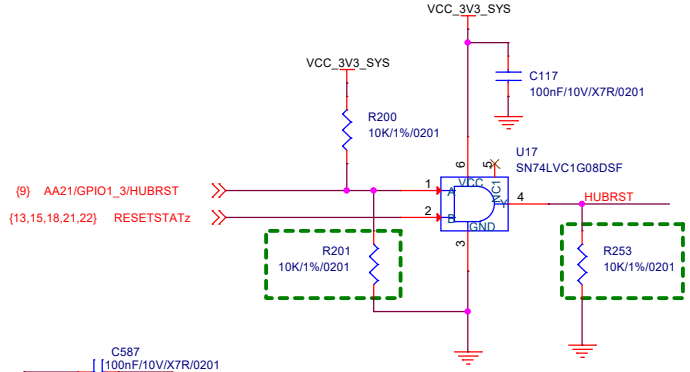
USB1 - USB 2.0 TYPE-A

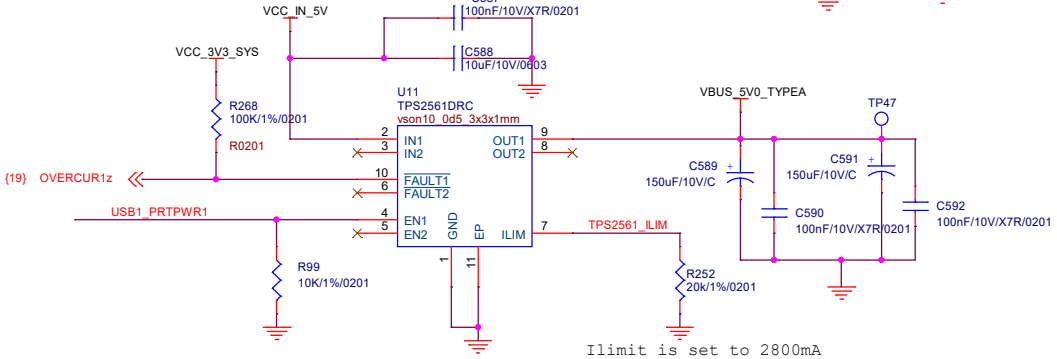


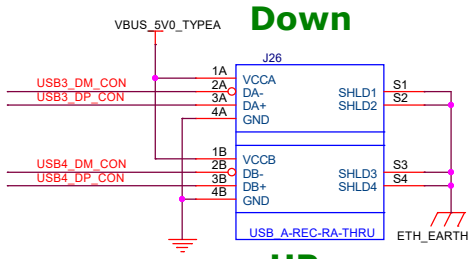


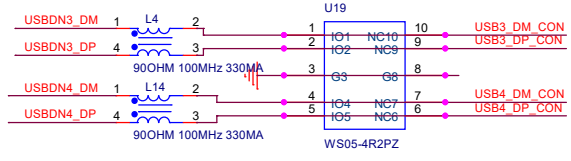


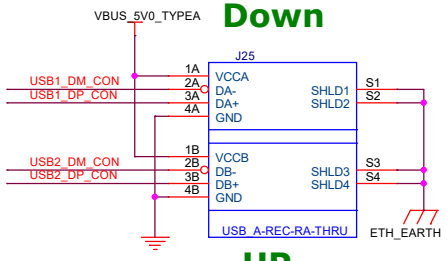


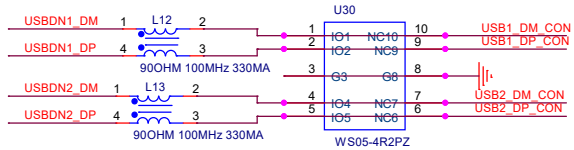








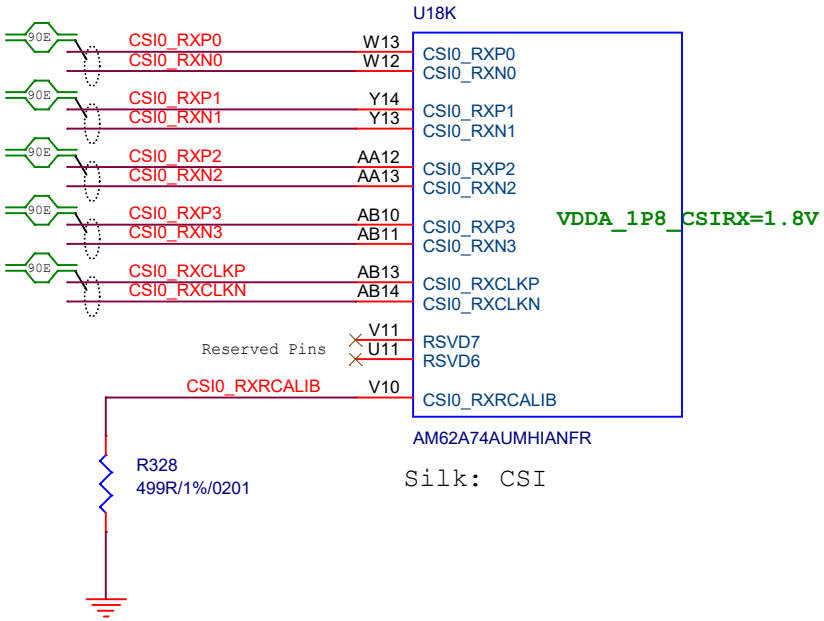




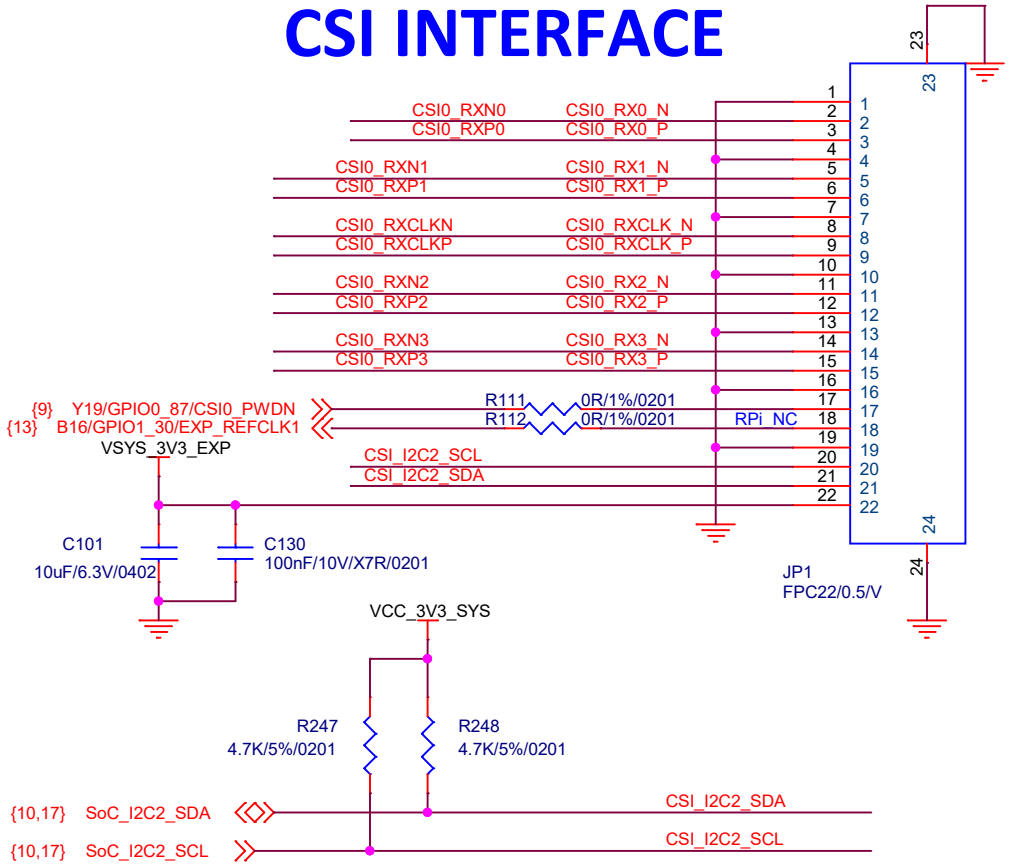
<Core Design>

inhand	Beijing InHand Networks Technology Co., Ltd.		
Project:	Mo_62A		
File:	19. USB1 HUB 4xTYPE-A		
Date:	Sunday, April 26, 2026	Rev:	V1.0
Designed by:	Reviewed by:	Sheet:	19 of 22

SOC CSI INTERFACE



CSI INTERFACE



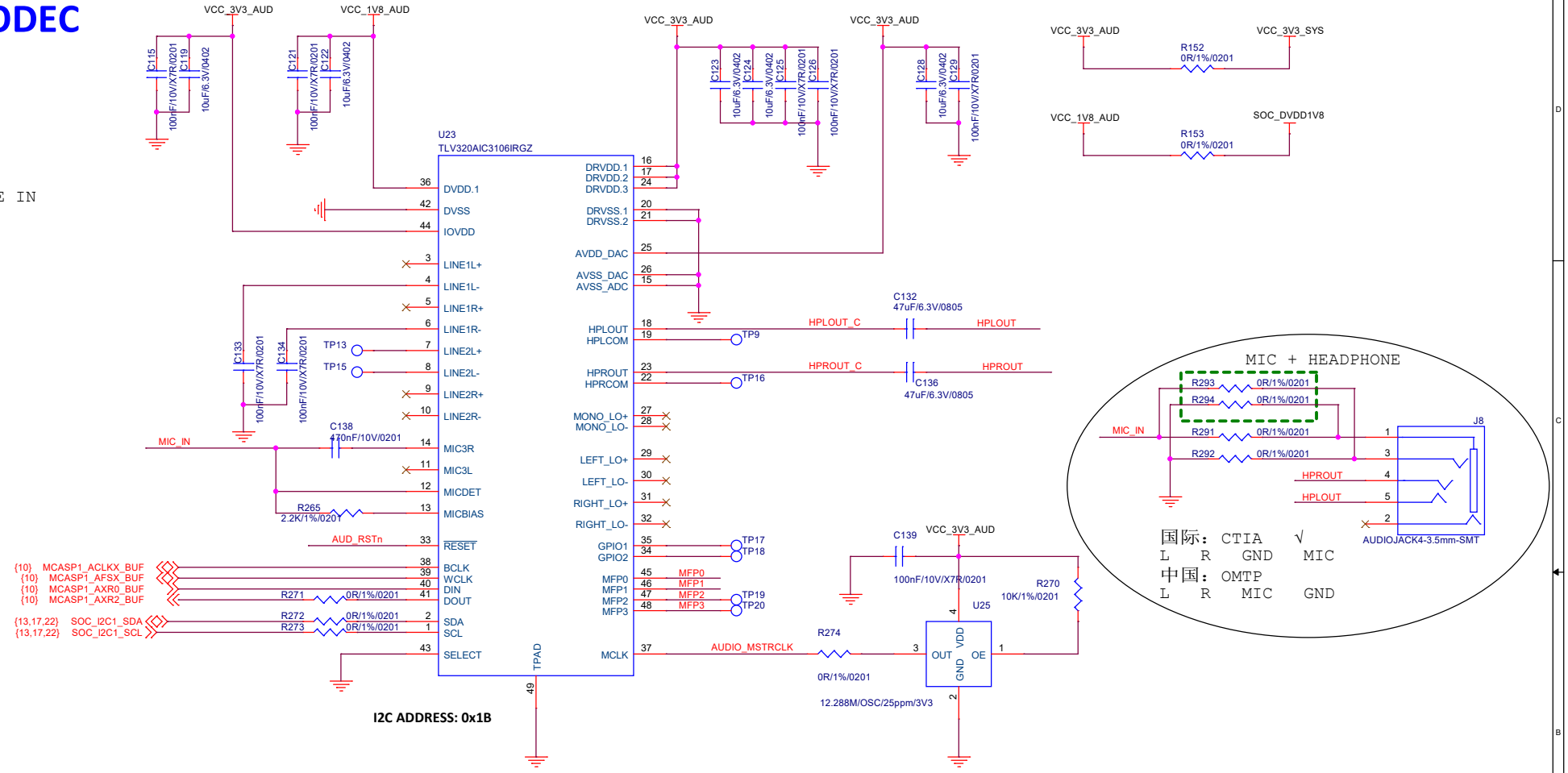
Raspberry Pi Camera Connector x4 Lane

<Core Design>

inhand	Beijing InHand Networks Technology Co., Ltd.		
Project:	Mo_62A		
File:	20. CSI INTERFACE		
Date:	Sunday, April 26, 2026	Rev:	V1.0
Designed by:	Reviewed by:	Sheet:	20 of 22

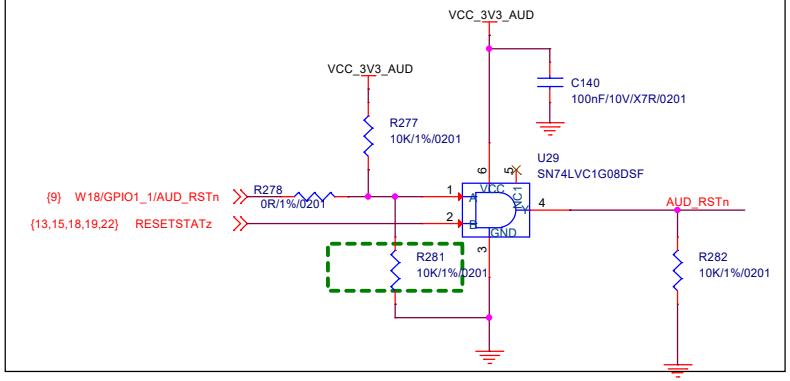
AUDIO CODEC

Silk: LINE IN



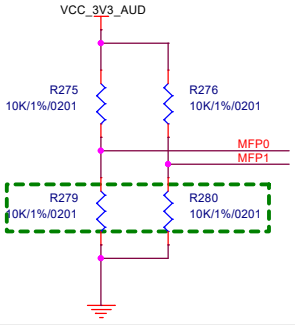
I2C ADDRESS: 0x1B

AUDIO CODEC RESET



CODEC I2C ADDRESS SELECTION

MFP0	MFP1	Device Address
0	0	0x18
0	1	0x19
1	0	0x1A
1	1	0x1B



<Core Design>

inhand	Beijing InHand Networks Technology Co., Ltd.		
Project:	Mo_62A		
File:	21. AUDIO CODEC		
Date:	Monday, April 27, 2026	Rev:	V1.0
Designed by:	Reviewed by:	Sheet:	21 of 22

HDMI INTERFACE

